

STL-FM: Trajectory Planning Through STL-Guided Flow Matching

Imposing temporal requirements on a generative motion prior at inference time

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Master's thesis defence, carried out at ABB Robotics, Västerås, Sweden
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Agenda

1 Motivation

2 Background

3 Method

4 Results

5 Conclusions

6 Future work

Motivation

Adhesive bonding with an ABB YuMi

A single-arm YuMi applies a bead of fast-setting glue to a prototype part.

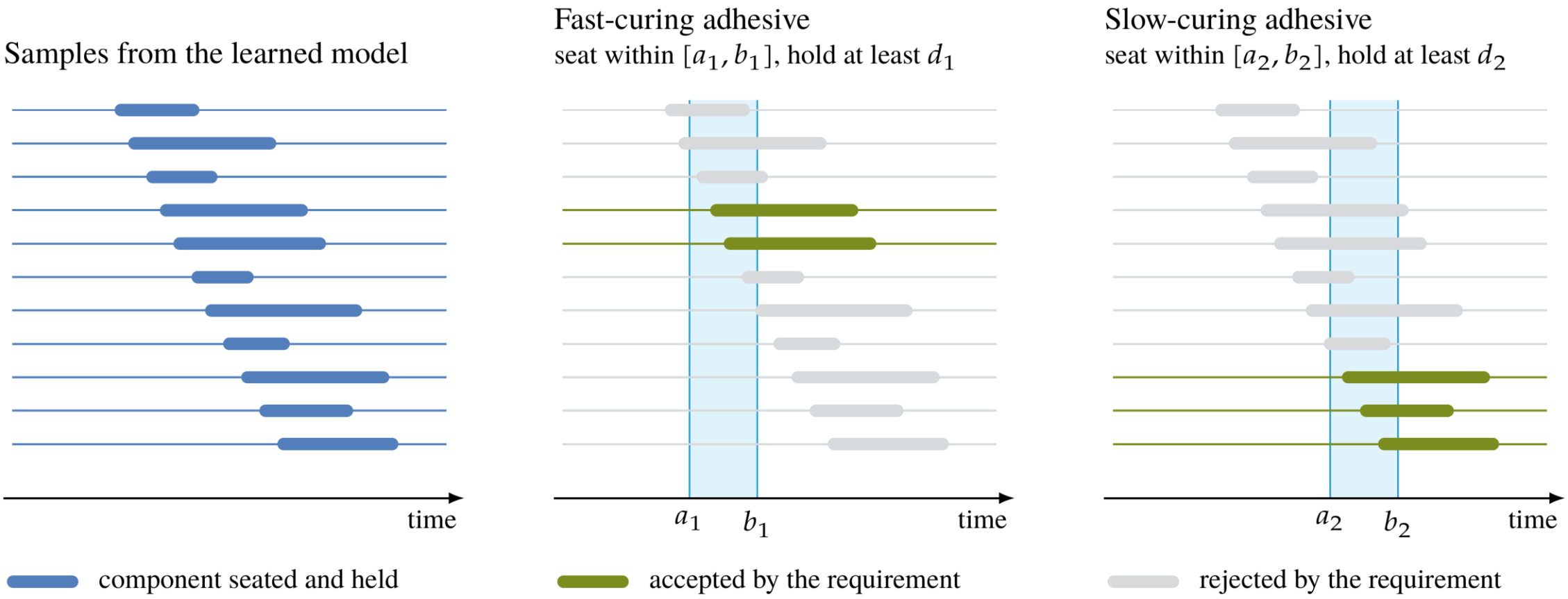
The part changes on a rapid schedule,
so the program changes with it.

“Even in half an hour, we can get some major changes done to the code.”

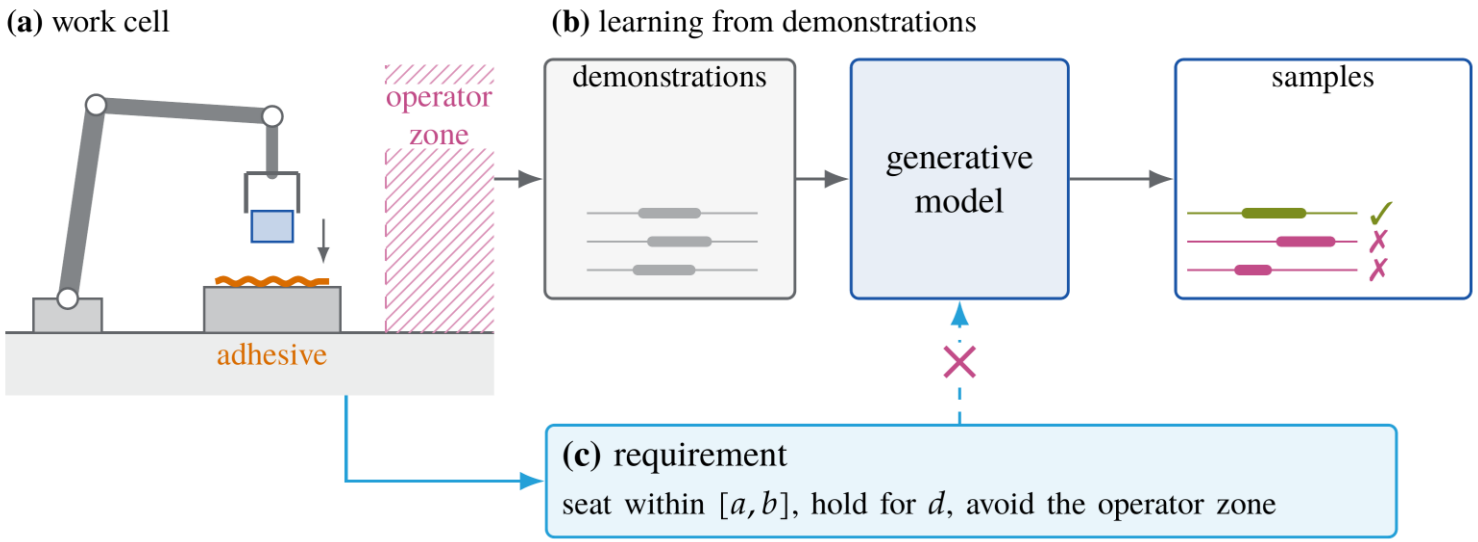


Photo and quote: ABB customer story,
GE Healthcare, Helsinki

Requirements that change after training



Problem



Given demonstrations recorded before training and a requirement φ stated after it, generate a trajectory that is plausible under the demonstrations and satisfies φ, without retraining.

Aim and research questions

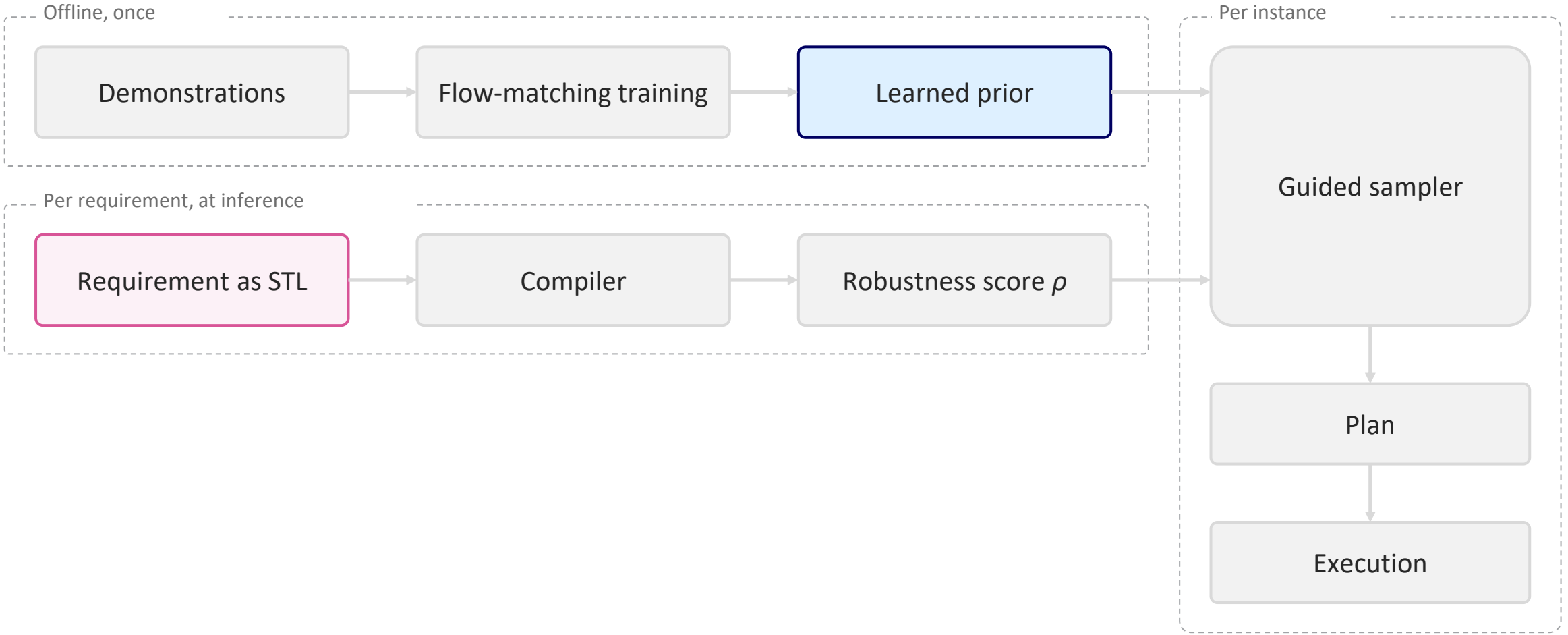
Aim

A method for imposing STL requirements on a trained generative model at inference time

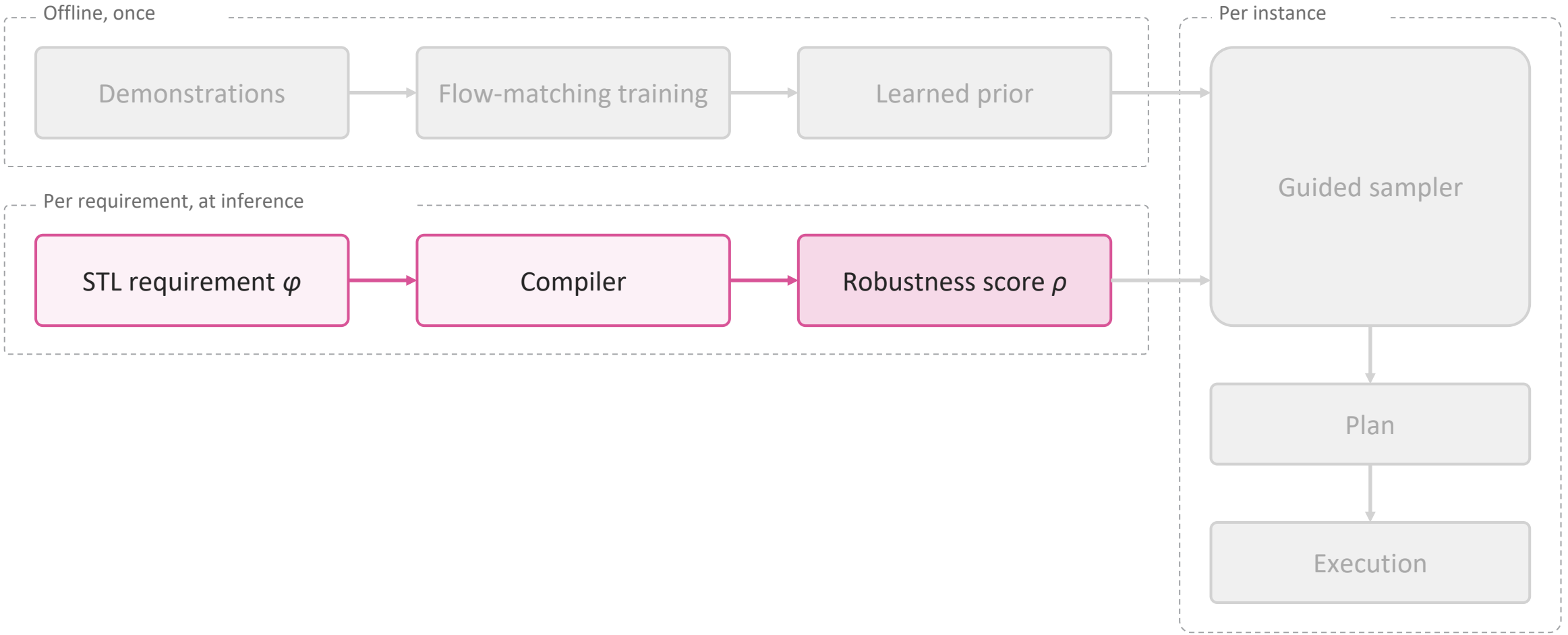
The trained model is called the learned prior, because it provides the motion that the requirement does not specify

RQ1	What does a demonstration witnessing a requirement imply about that requirement?
RQ2	What does maximizing the robustness score alone determine about a trajectory?
Method	STL-FM, robustness guidance of the learned prior

Approach


Offline, once
Demonstrations
Flow-matching training
Learned prior
Per requirement, at inference
Requirement as STL
Compiler
Robustness score ρ
Per instance
Guided sampler
Plan
Execution

Stating the requirement



Signal Temporal Logic

Seat within [a, b], hold for d, avoid the operator zone

$$\varphi = \Diamond_{[a,b]} \Box_{[0,d]} \, seated \wedge \Box_{[0,T]} \neg \, operator$$

Evaluated fragments: predicates μ, negation, conjunction, bounded ◇ and □

Maler and Nickovic 2004

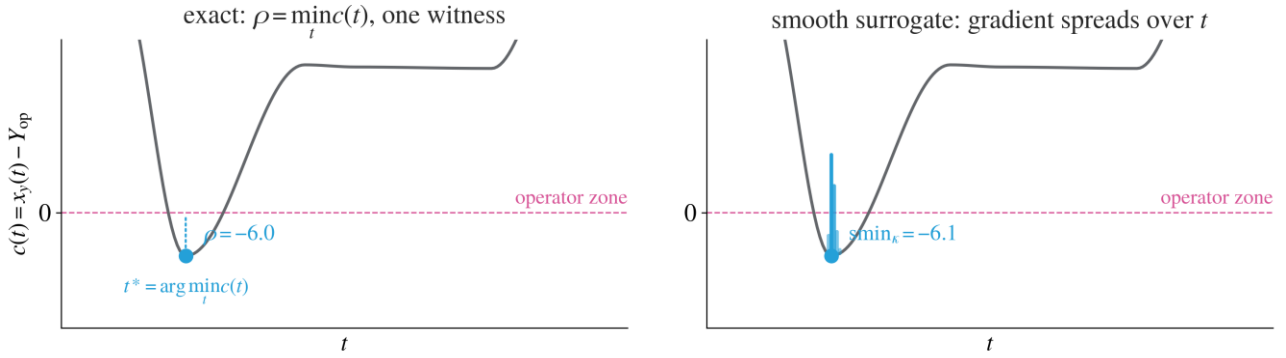
Robustness

$$\rho_\varphi(\tau) > 0 \implies \tau \models \varphi$$

$$\rho(\mu, x, t) = h(x(t))$$
$$\rho(\neg\varphi, x, t) = -\rho(\varphi, x, t)$$
$$\rho(\varphi_1 \wedge \varphi_2, x, t) = \min\left(\rho(\varphi_1, x, t), \rho(\varphi_2, x, t)\right)$$

$$\rho(\Diamond_{[a,b]}\varphi, x, t) = \max_{t' \in [t+a, t+b]} \rho(\varphi, x, t')$$
$$\rho(\Box_{[a,b]}\varphi, x, t) = \min_{t' \in [t+a, t+b]} \rho(\varphi, x, t')$$

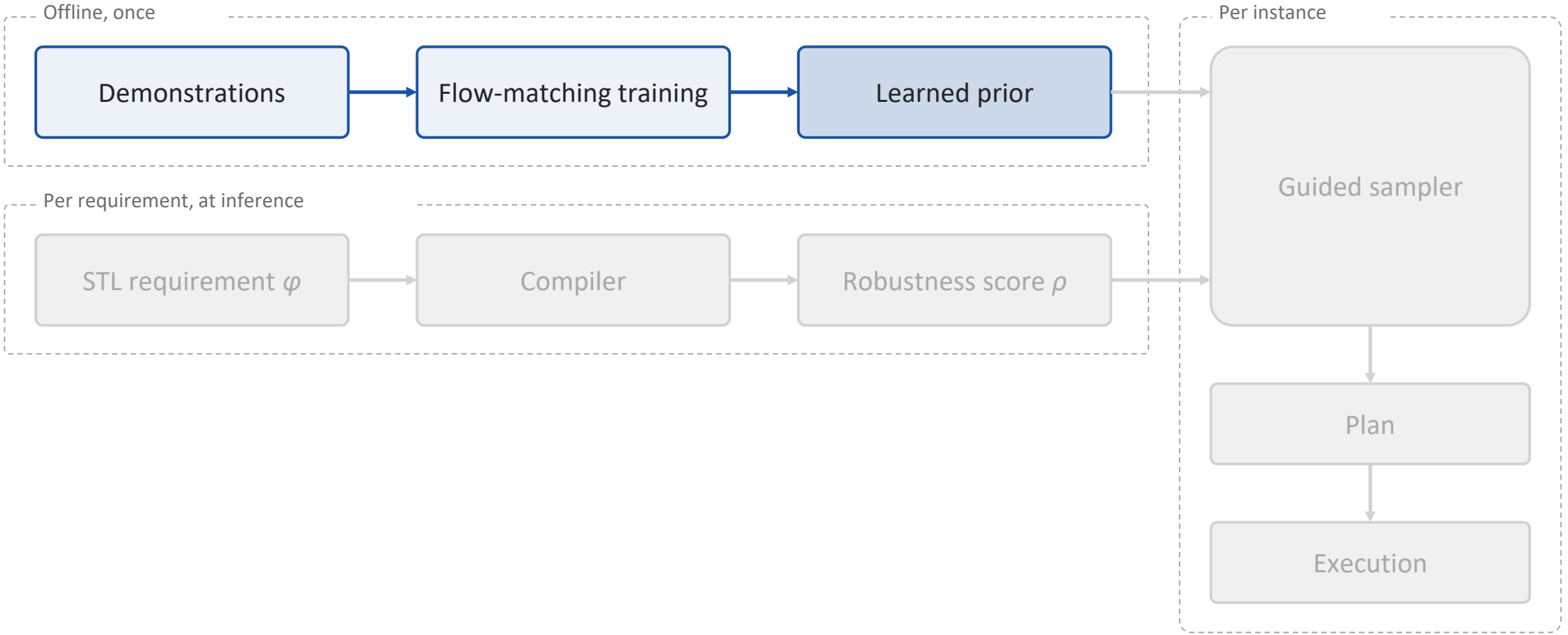
$$\mathrm{smax}_\kappa(z_1, \ldots, z_n) = \frac{1}{\kappa} \log \sum_i e^{\kappa z_i},$$
$$\mathrm{smin}_\kappa(z) = -\,\mathrm{smax}_\kappa(-z)$$



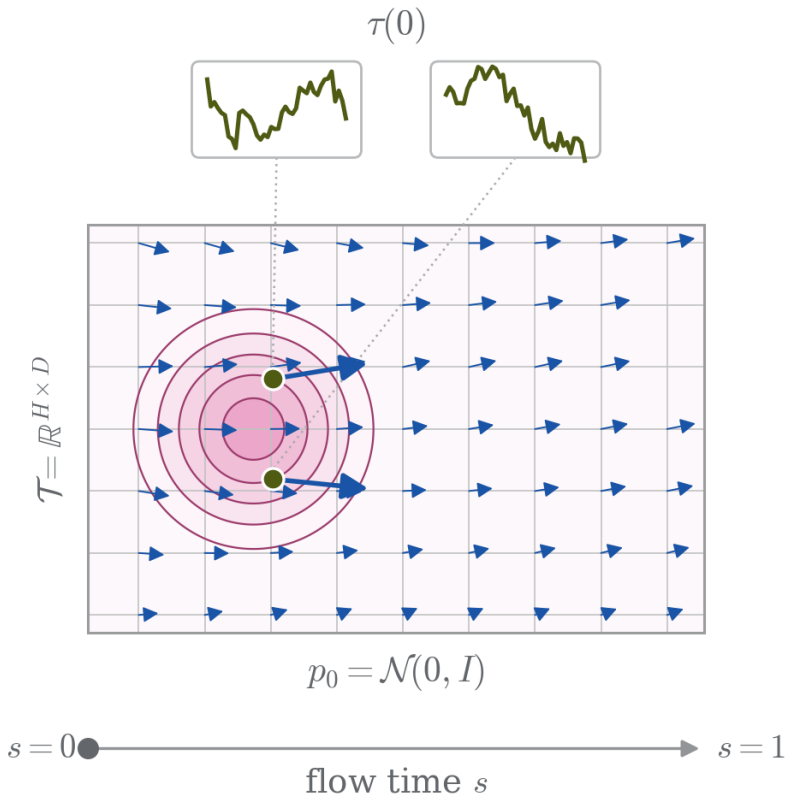
Guidance uses the surrogate at κ = 10. Satisfaction is always scored with the exact semantics.

Donzé and Maler 2010; Leung et al. 2021; Kapoor et al. 2025 (STLCG++)

Learning the prior



Flow matching



Lipman et al. 2023; Liu et al. 2022

Training

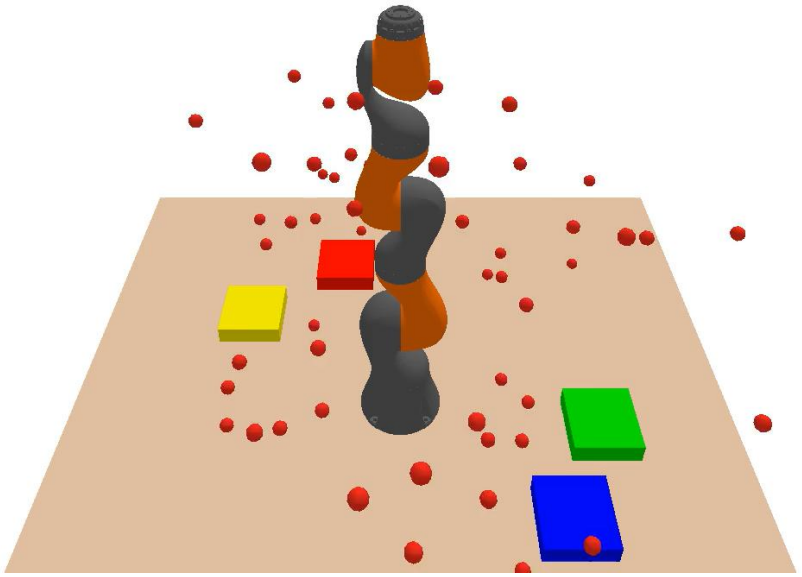
$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}\left[\|v_\theta(\psi_s, s) - (\tau_1 - \tau_0)\|^2\right]$$
$$\psi_s = (1-s)\,\tau_0 + s\,\tau_1, \qquad \tau_0 \sim \mathcal{N}(0, I), \quad \tau_1 \sim p_{\mathrm{data}}, \quad s \sim \mathcal{U}[0,1]$$

Sampling

$$\frac{d\tau}{ds} = v_\theta(\tau, s), \qquad \tau(0) \sim \mathcal{N}(0, I), \qquad s : 0 \to 1$$

K = 64 Euler steps. The prior is unconditional, trained once on demonstrations, and its only conditioning is the start state.

Generation from noise

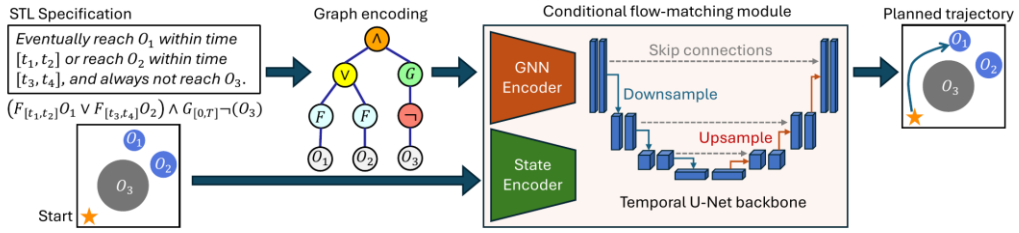


Janner et al. 2022

Related work

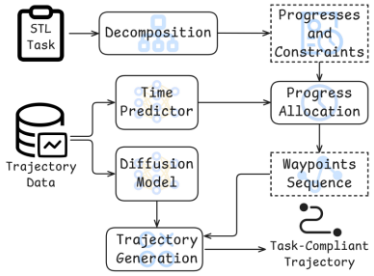
Training-time conditioning

TeLoGraF, Meng and Fan 2025



Decompose-and-stitch

ZSTP, Liu et al. 2025



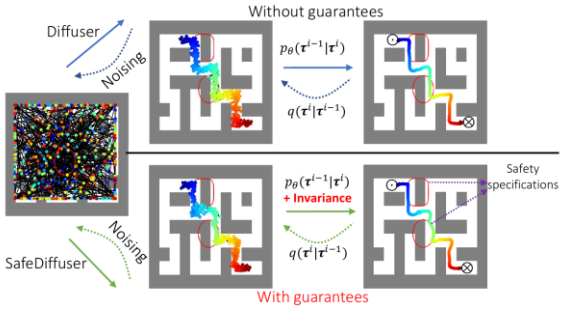
Trajectory optimization

Kurtz and Lin 2021



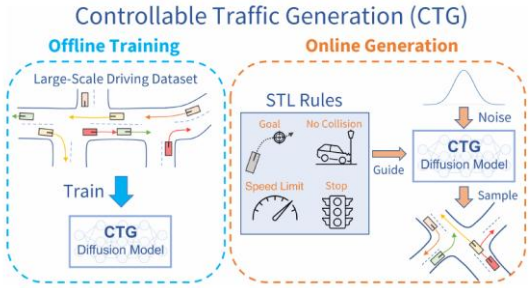
Safe-set enforcement

SafeDiffuser, Xiao et al. 2023



Inference-time robustness guidance

CTG, Zhong et al. 2022



STL-FM
on a flow-matching prior

Overview figures reproduced from the cited papers

Guiding the prior



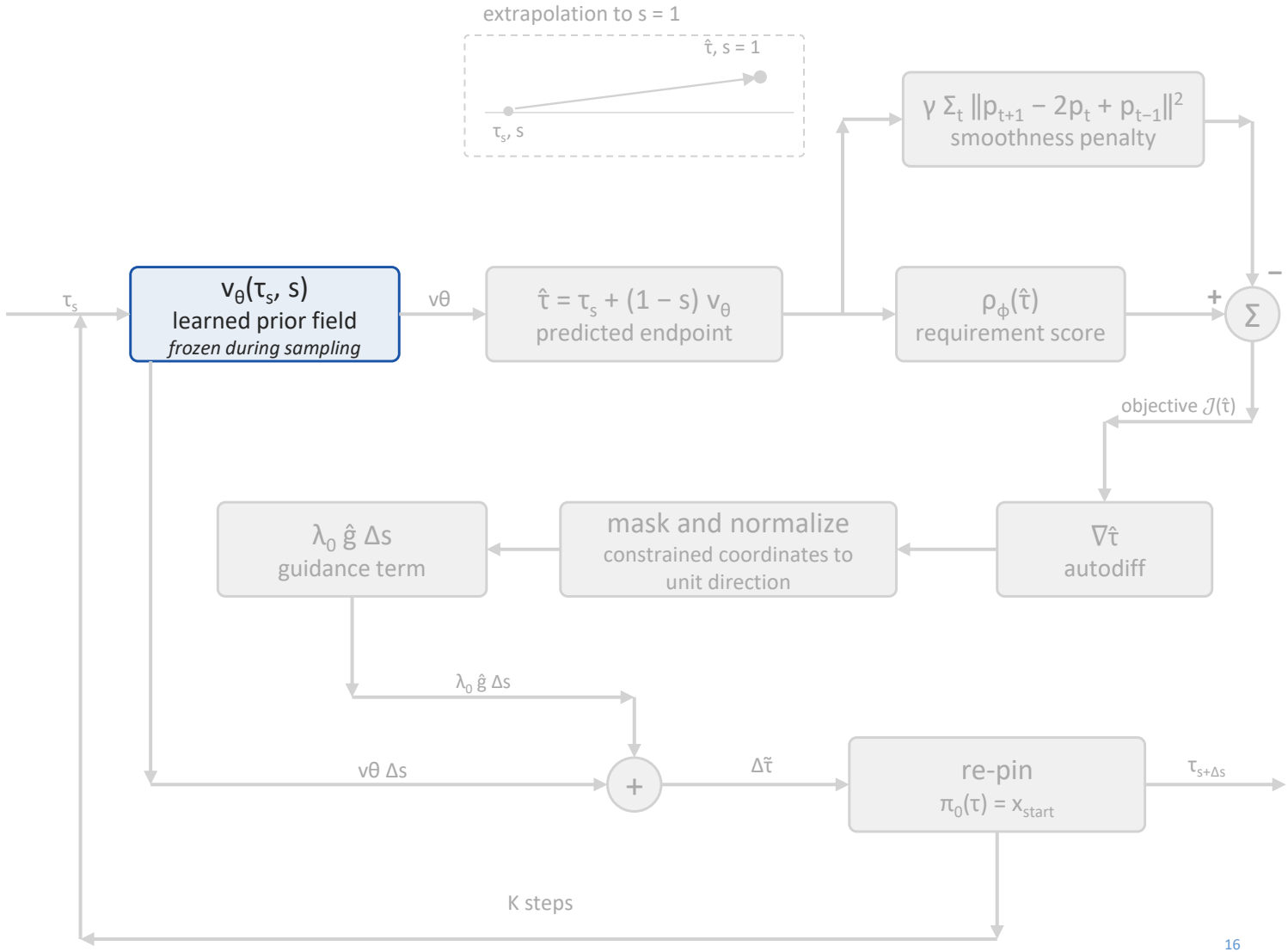
Guided velocity field

$$v_{\text{guided}}(\tau, s) = \underbrace{v_\theta(\tau, s)}_{\text{learned prior}} + \lambda_0 \underbrace{\tilde{\nabla}_{\hat{\tau}} \mathcal{J}(\hat{\tau}(s))}_{\substack{\text{gradient of the objective,} \\ \text{at the predicted endpoint}}}$$

$$\mathcal{J}(\hat{\tau}) = \rho_\varphi(\hat{\tau}) - \gamma \sum_t \|p_{t+1} - 2p_t + p_{t-1}\|^2$$

$$\tau_{s+\Delta s} = \tau_s + \Delta s \, v_{\text{guided}}(\tau_s, s)$$

$$\pi_0(\tau_{s+\Delta s}) = x_{\text{start}}$$



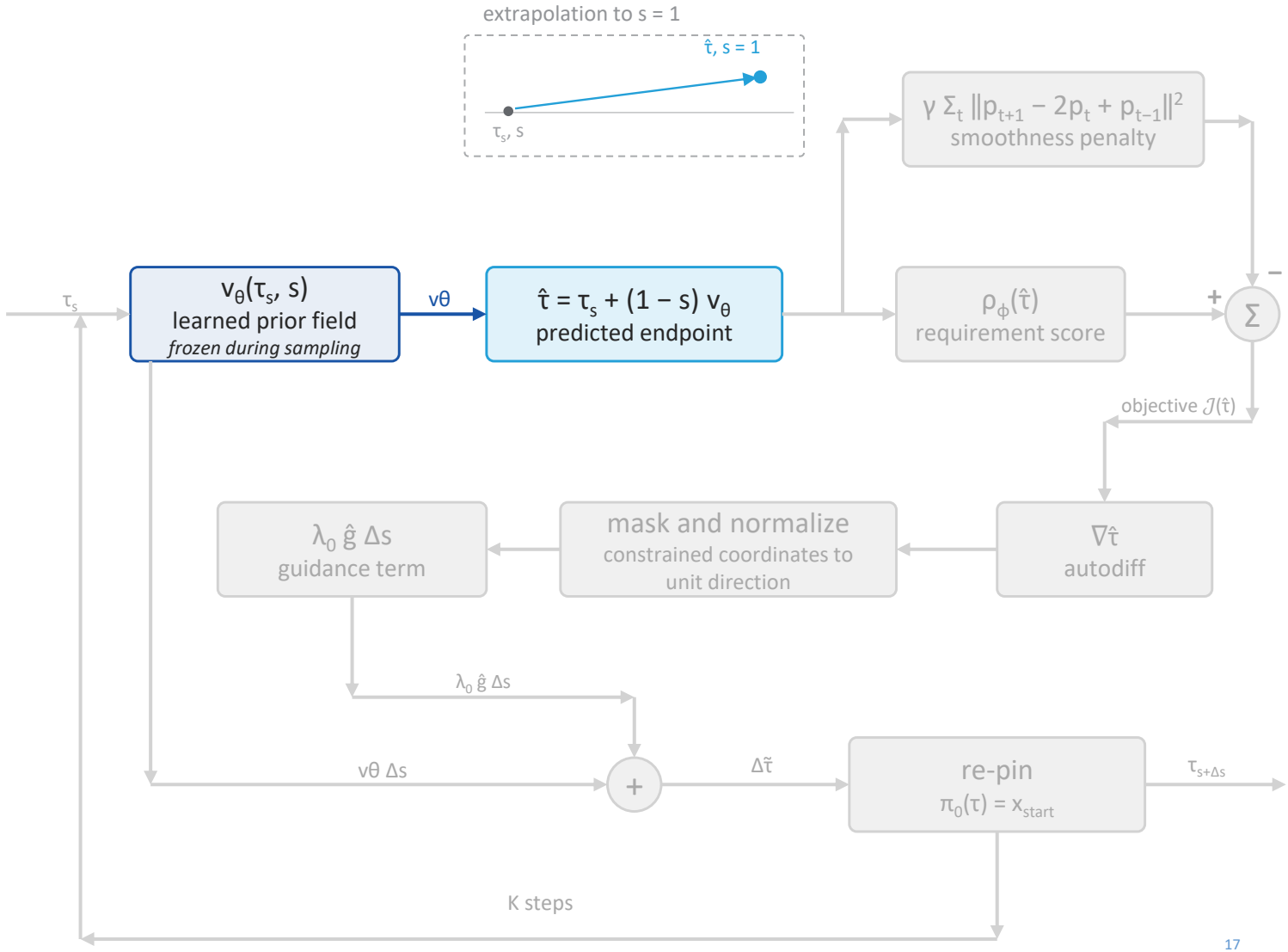
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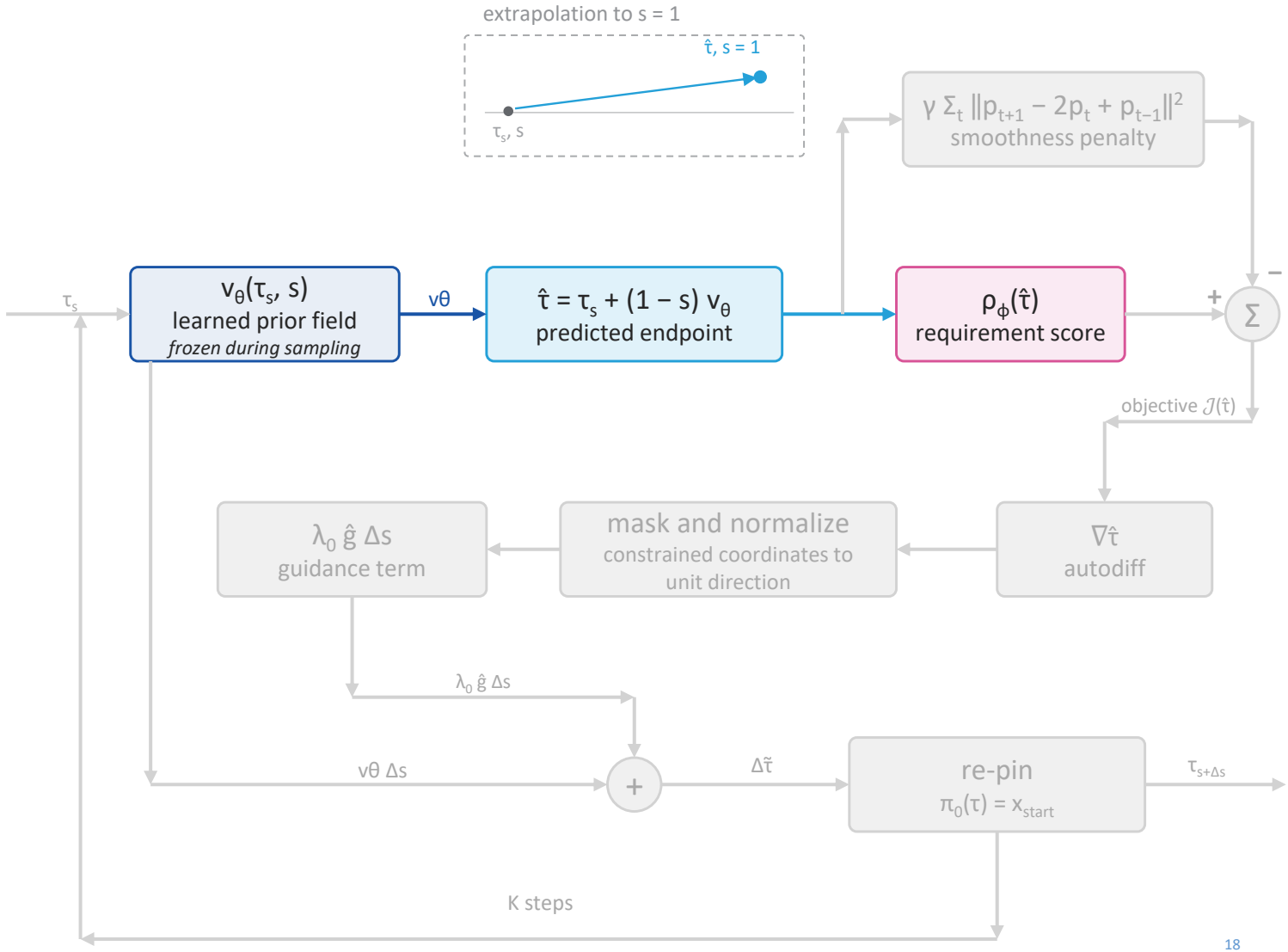
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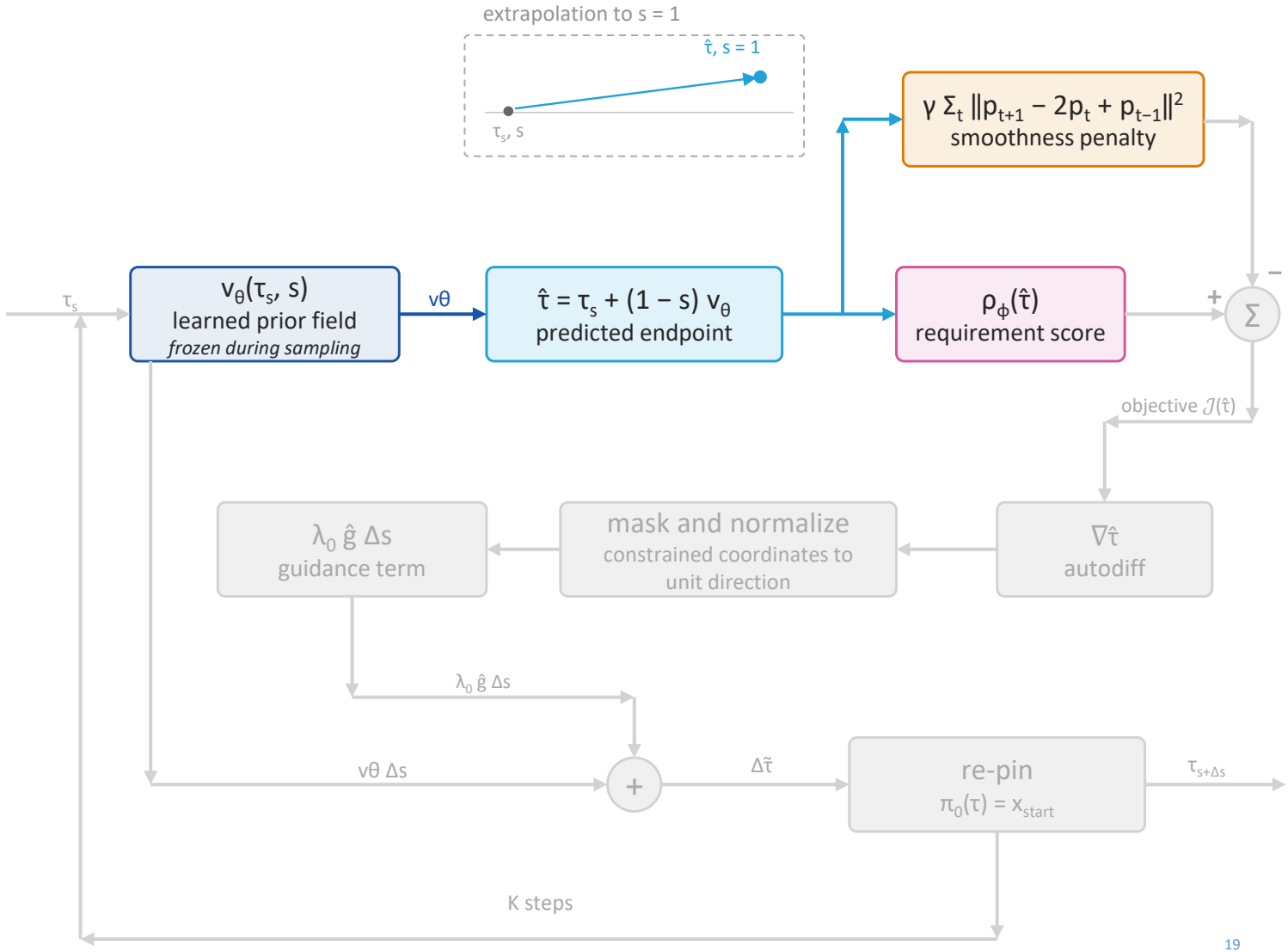
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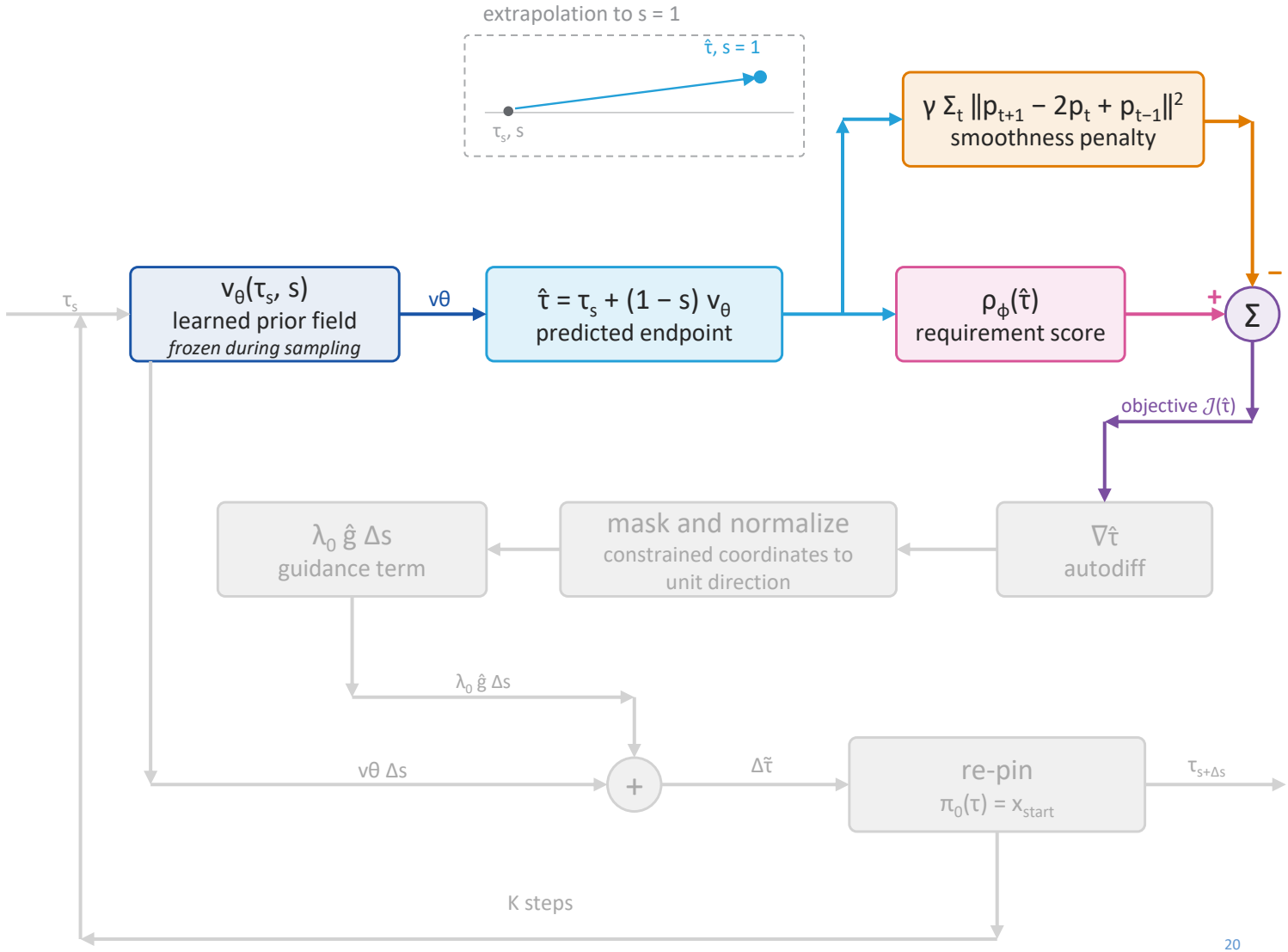
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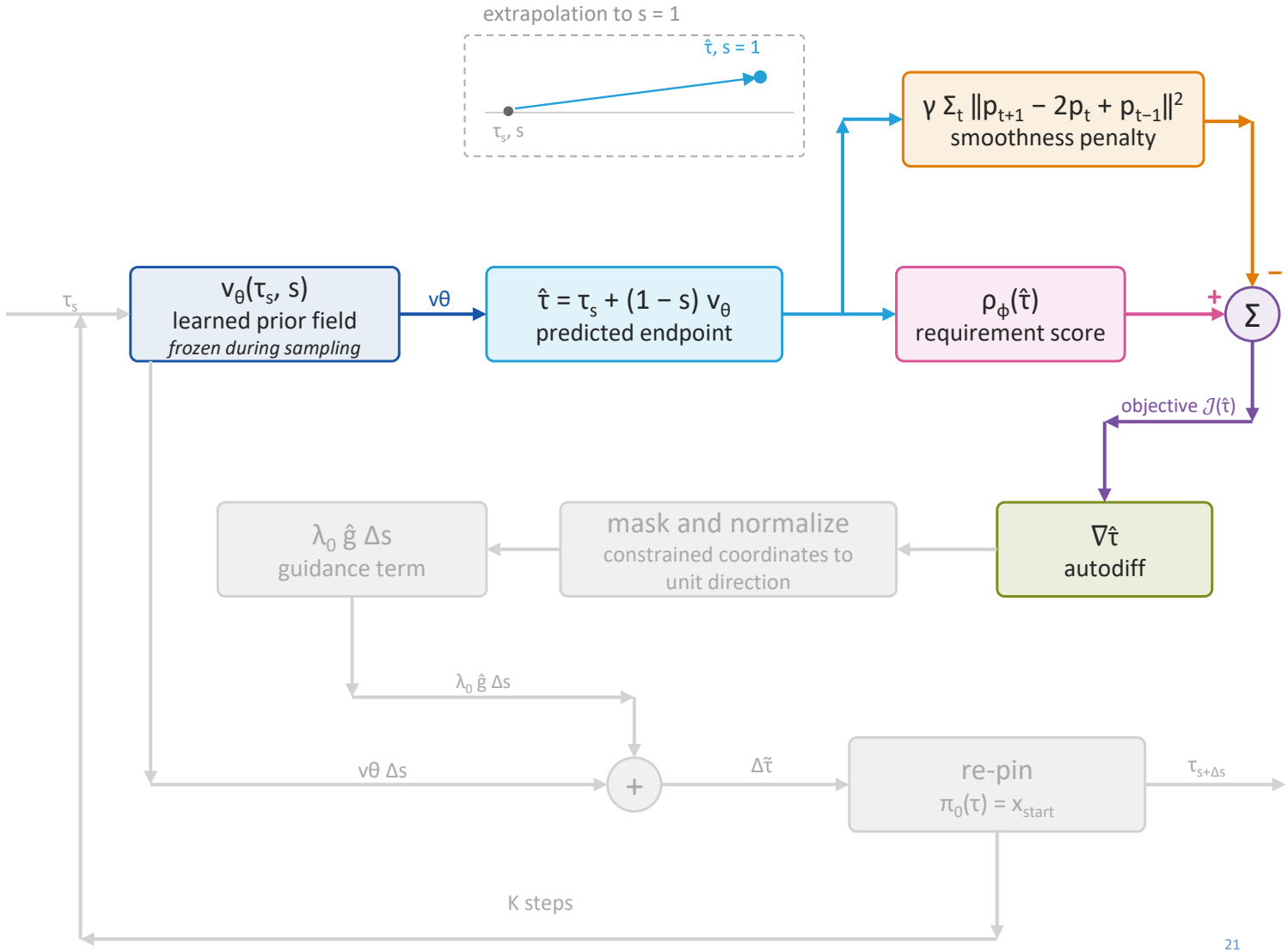
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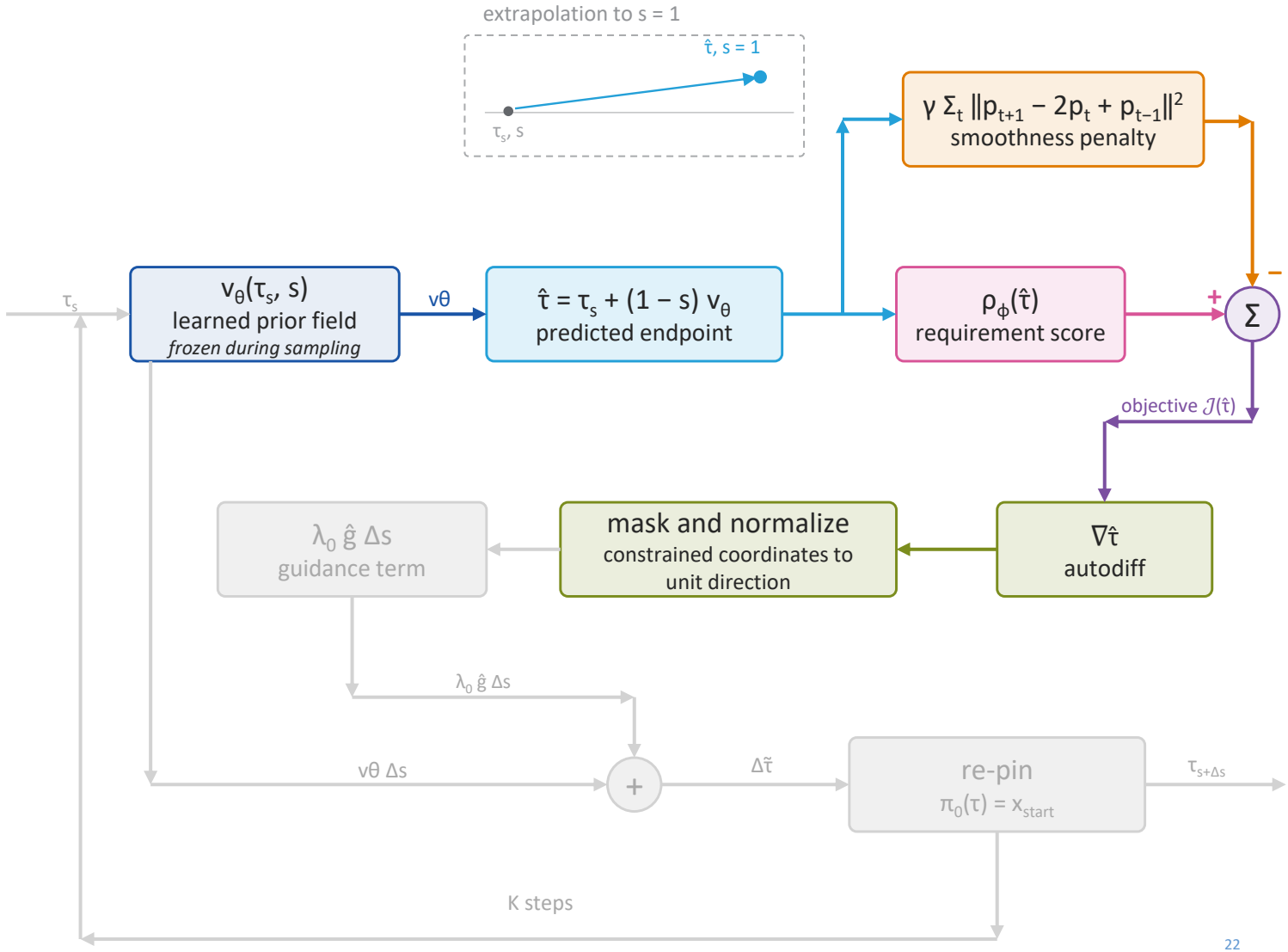
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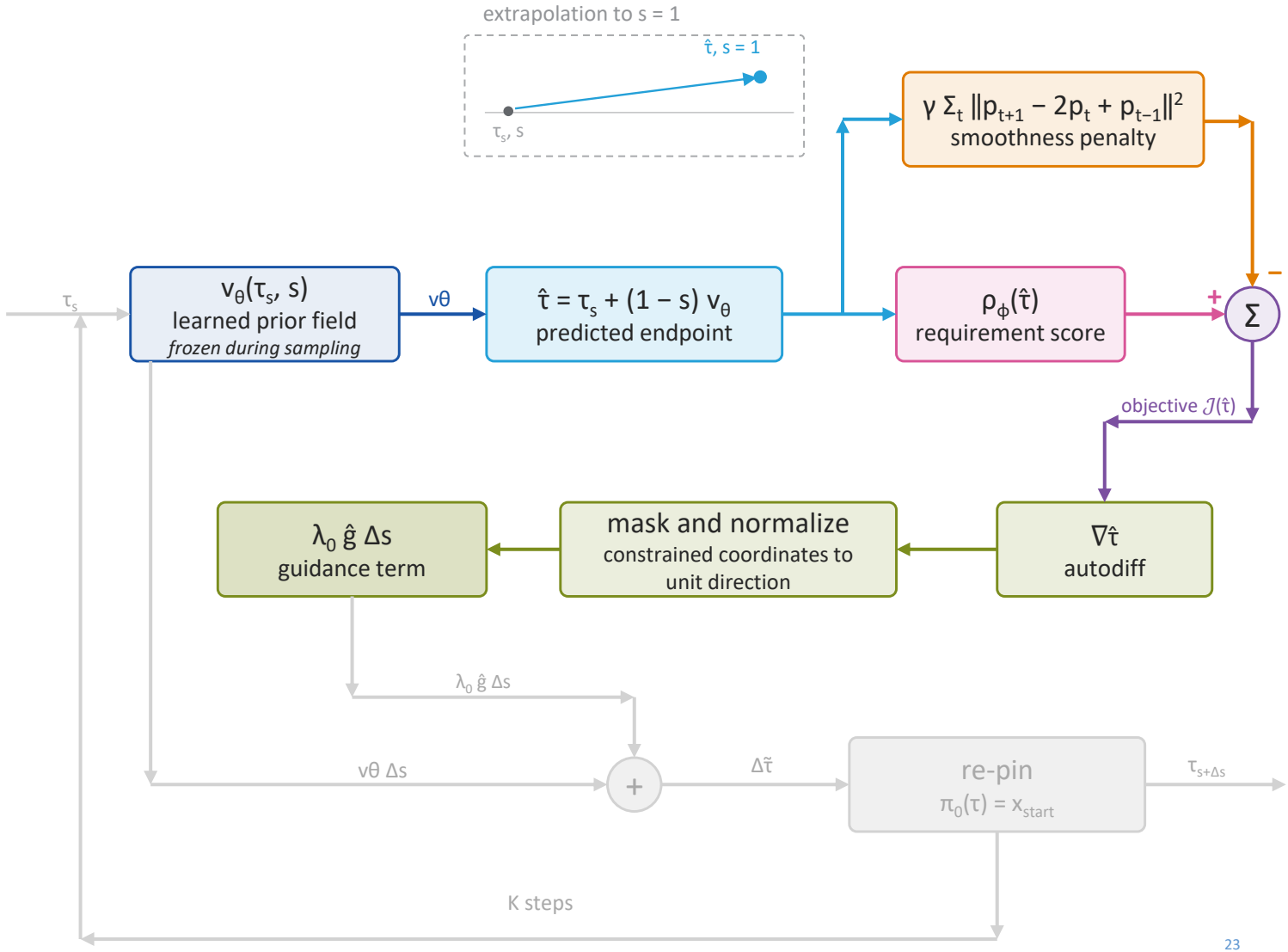
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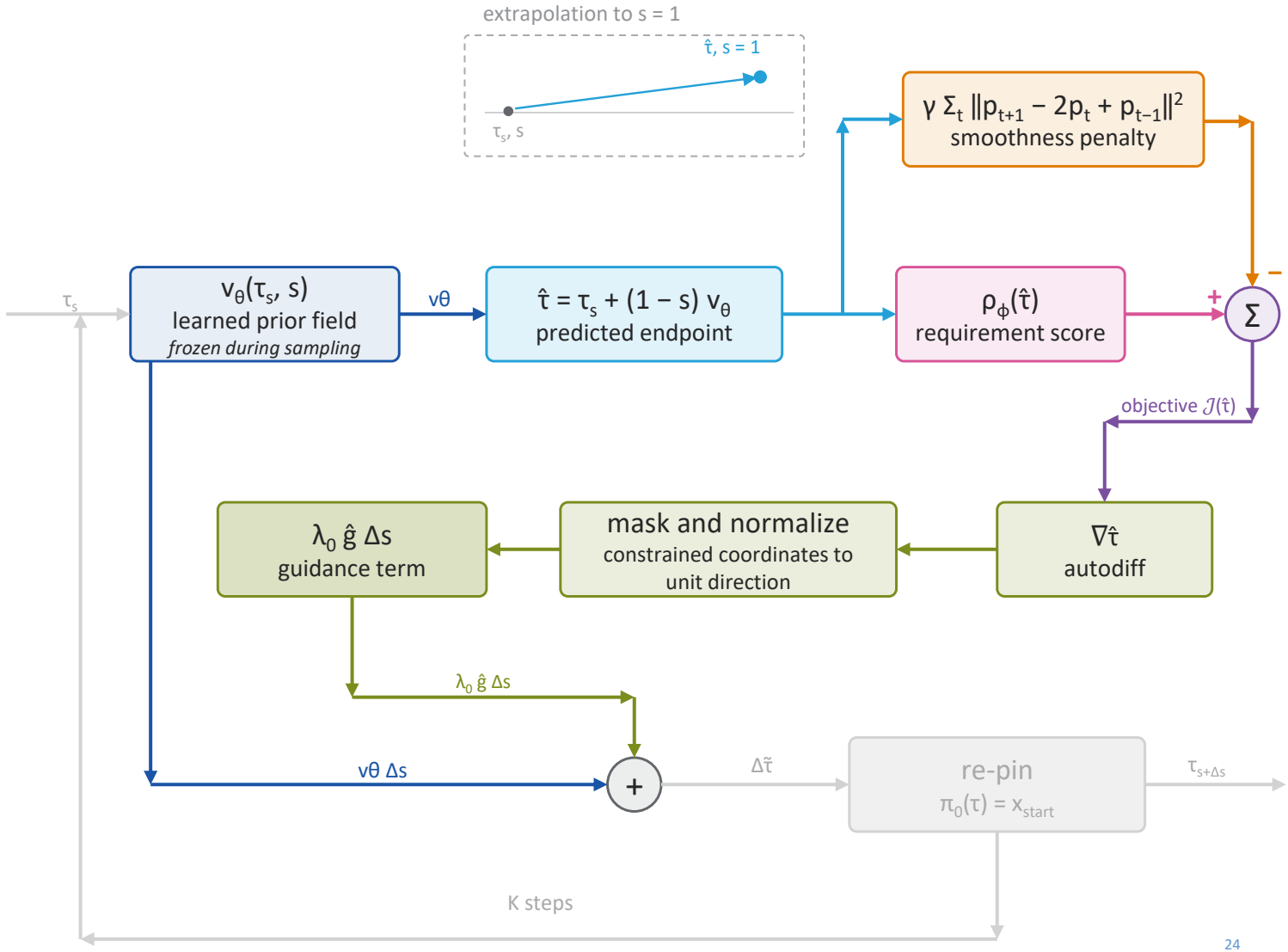
Guided sampler

$$v_{\text{guided}}(\tau, s) = \underbrace{v_\theta(\tau, s)}_{\text{learned prior}} + \lambda_0 \underbrace{\tilde{\nabla}_{\hat{\tau}} \mathcal{J}(\hat{\tau}(s))}_{\substack{\text{gradient of the objective,} \\ \text{at the predicted endpoint}}}$$

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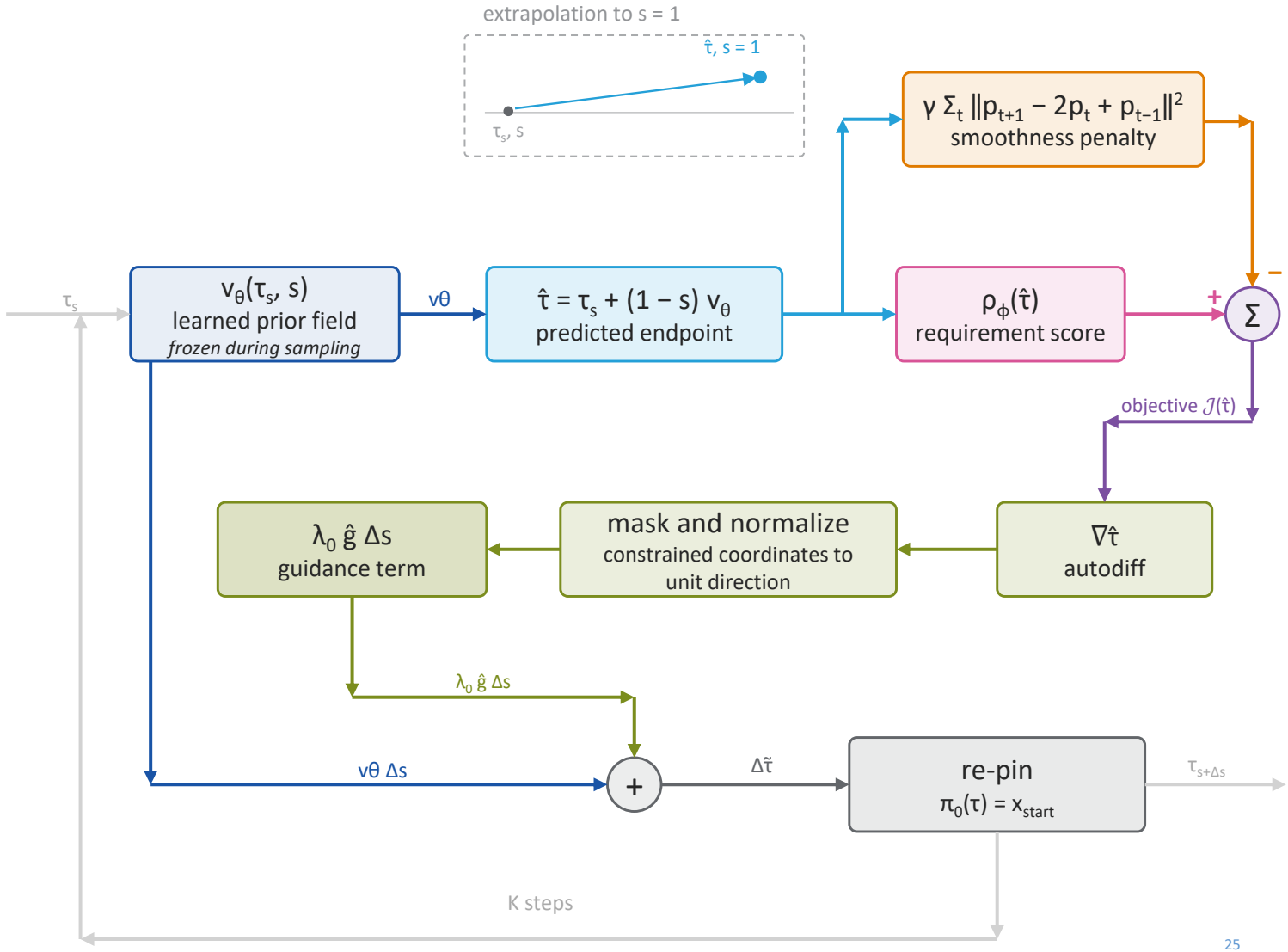
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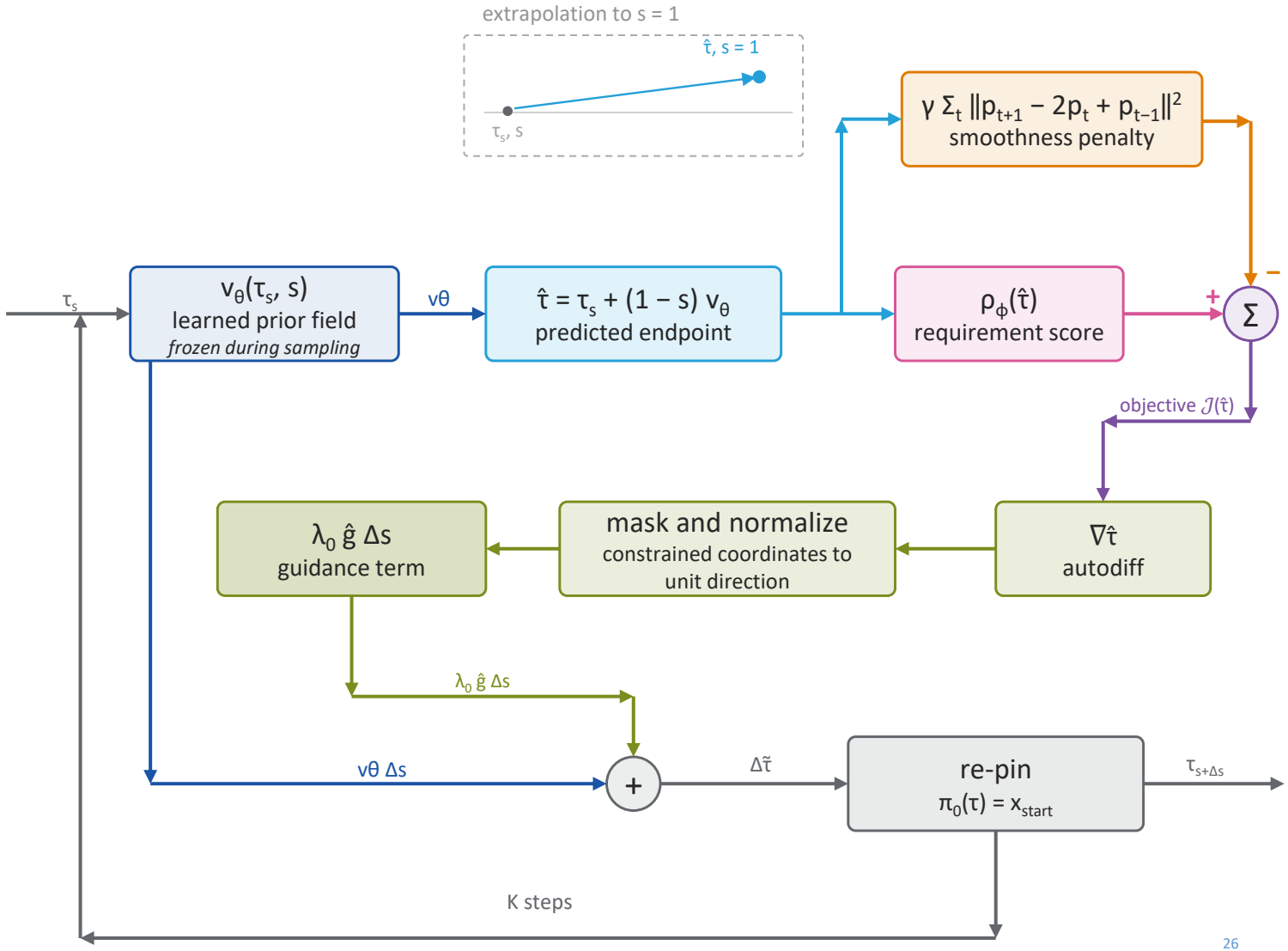
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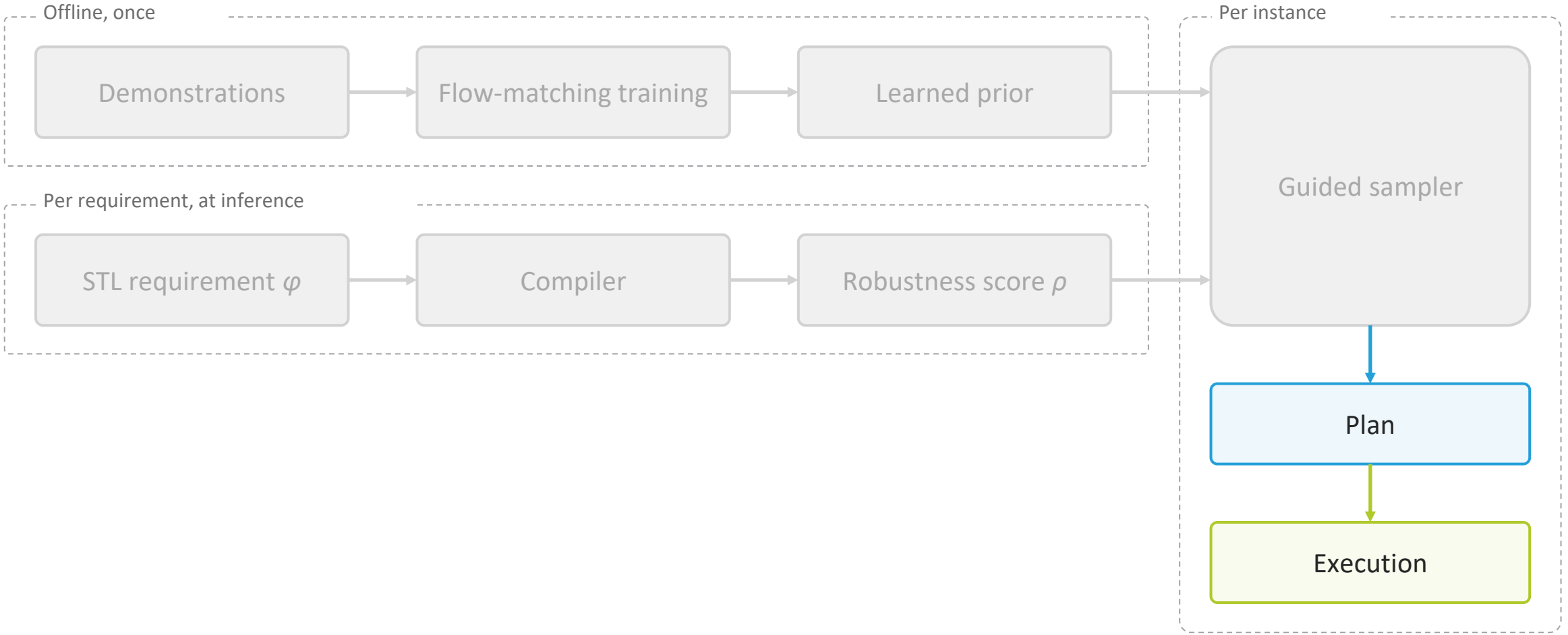
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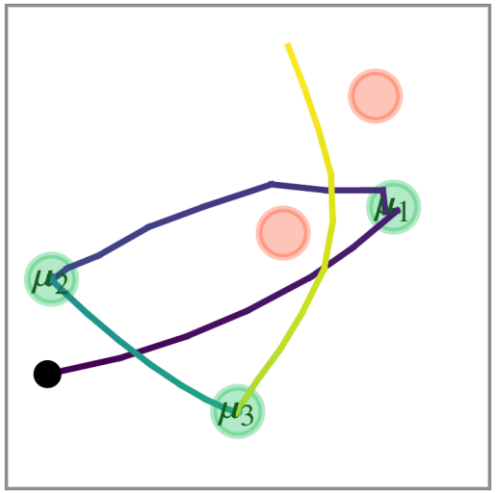
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Evaluating the plans

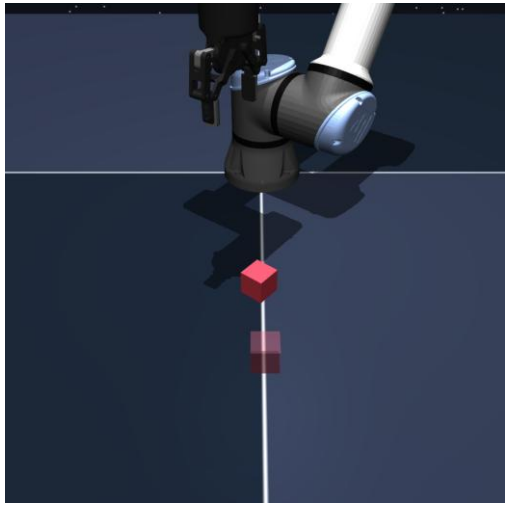


Experimental setup



Planar navigation

100 runs per class, plans only



Cube manipulation

10 runs per class, executed

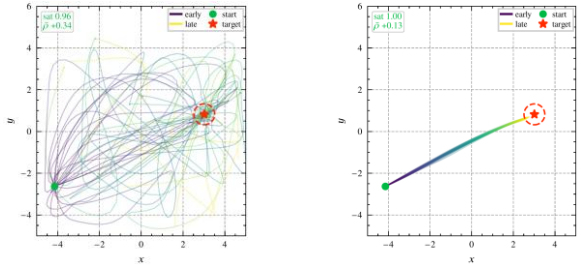
Planner	Learned prior	Requirement	Endpoint pin
Endpoint-inpaint planner	✓		✓
STL-FM	✓	✓	
STL Optimizer (no-prior)		✓	

STL-FM with endpoint pin differs from STL-FM in satisfaction rate by at most 0.05 on any of the six classes (backup)

Six requirement classes

Reach

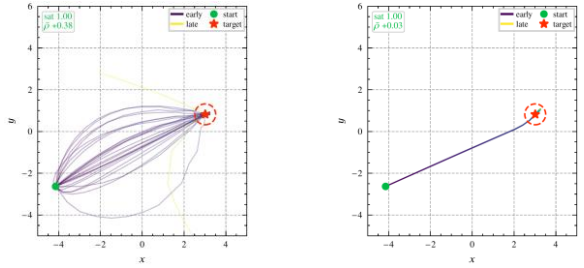
$\varphi = \Diamond_{[0,63]}\, \mu_1$



STL-FM | STL Optimizer

Nested reach-and-dwell

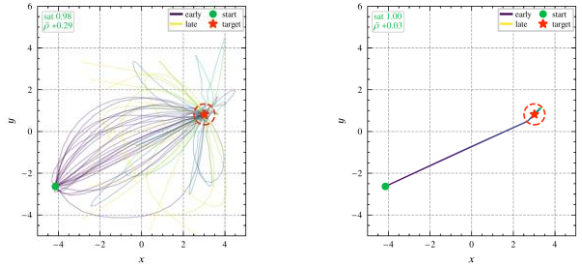
$\varphi = \Diamond_{[0,15]} \Box_{[0,48]}\, \mu_1$



STL-FM | STL Optimizer

Recurrence

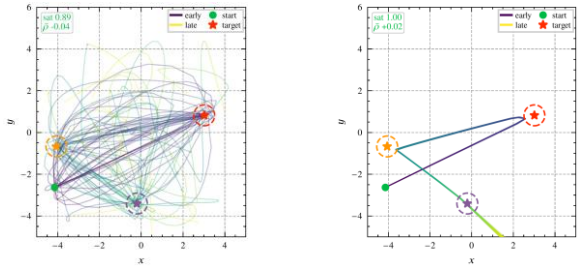
$\varphi = \Box_{[0,48]} \Diamond_{[0,15]}\, \mu_1$



STL-FM | STL Optimizer

Sequence

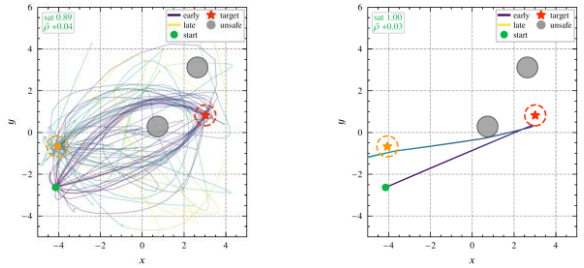
$\varphi = \Diamond_{[0,15]}\mu_1 \wedge \Diamond_{[15,30]}\mu_2 \wedge \Diamond_{[30,45]}\mu_3$



STL-FM | STL Optimizer

Sequence with avoidance

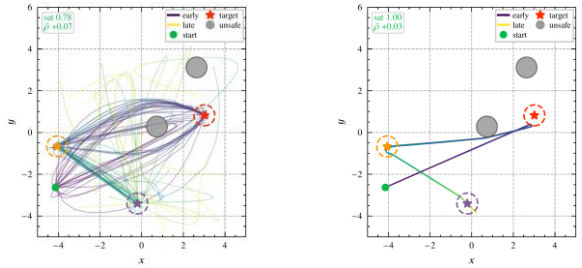
$\varphi = \Diamond_{[0,15]}\mu_1 \wedge \Diamond_{[15,30]}\mu_2 \wedge \Box_{[0,63]}\neg\mu_3 \wedge \Box_{[0,63]}\neg\mu_4$



STL-FM | STL Optimizer

Composite

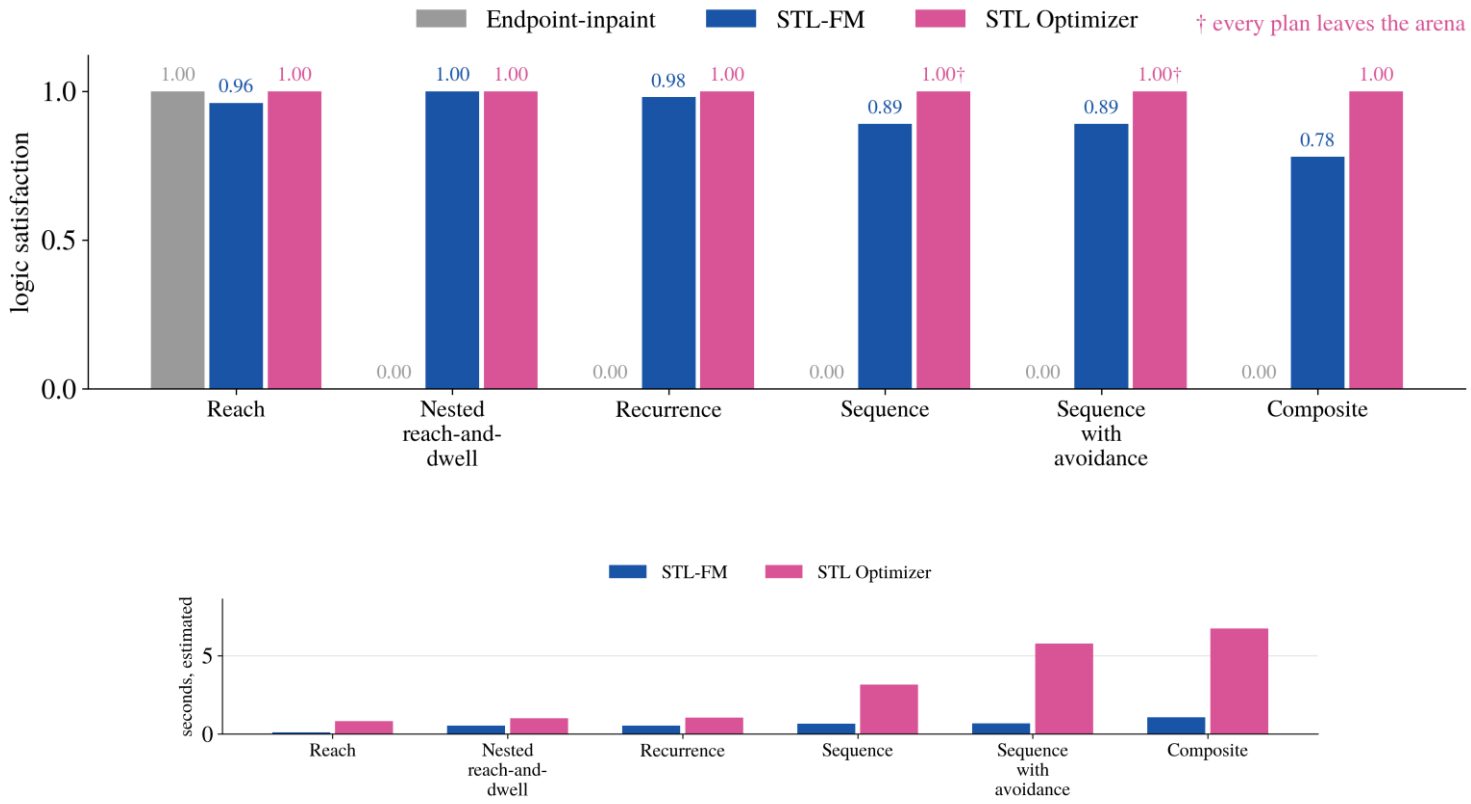
$\varphi = \Diamond_{[0,15]}(\mu_1 \wedge \Diamond_{[0,15]}(\Box_{[0,10]}\mu_2 \wedge \Diamond_{[10,22]}\Box_{[0,11]}\mu_3)) \wedge \Box_{[0,63]}(\neg\mu_4 \wedge \neg\mu_5)$



STL-FM | STL Optimizer

One learned prior and one constant strength λ_0 are used for all six requirement classes

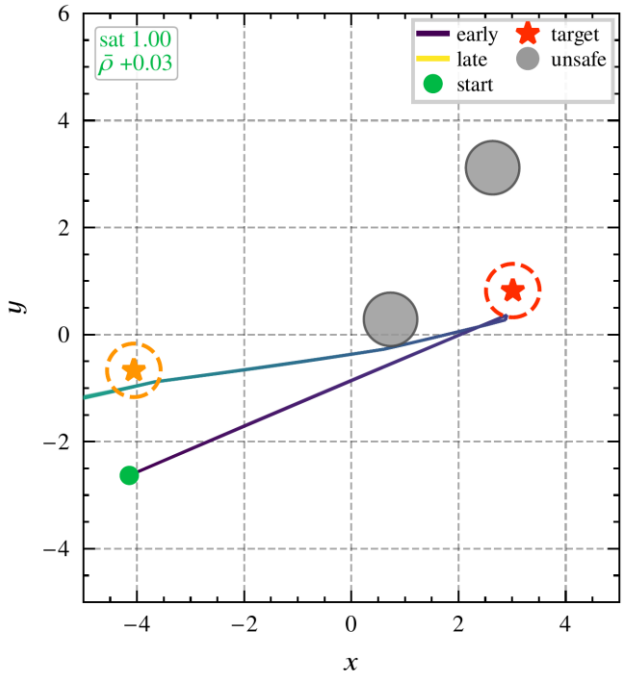
Logic satisfaction and time to satisfaction



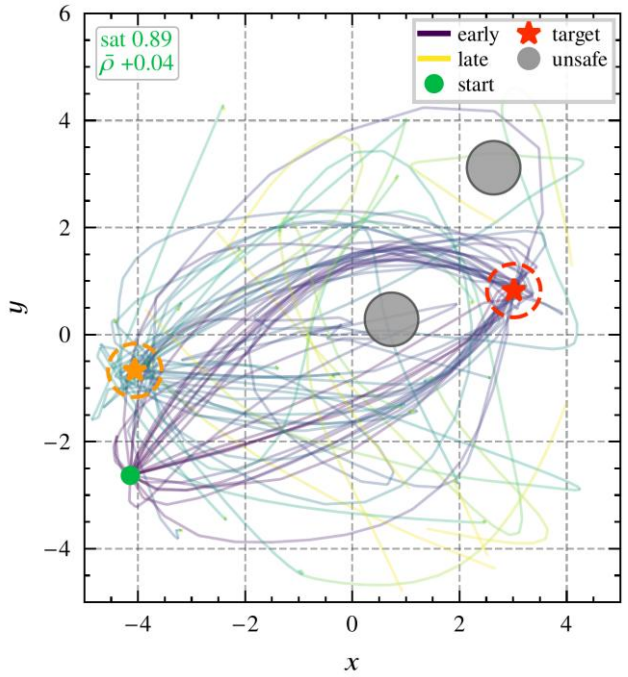
STL-FM satisfies most classes at comparable rates and plans 5 to 25 times faster

Arena excursion

STL Optimizer



STL-FM

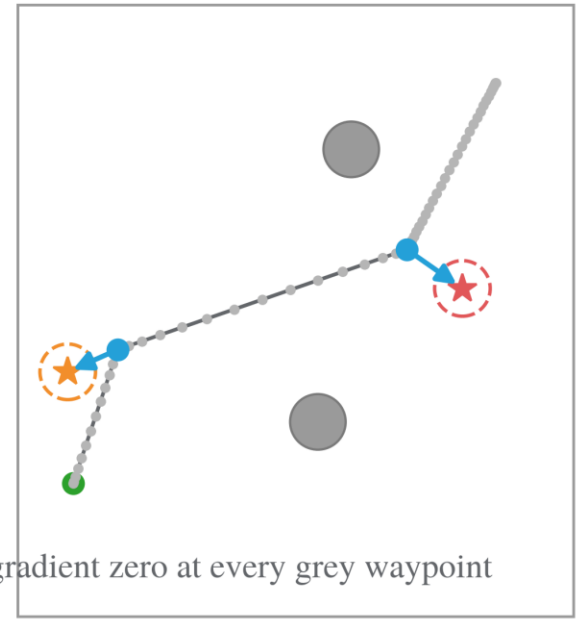


Sequence with avoidance, 25 of 100 plans per planner, arena clipped at its edge

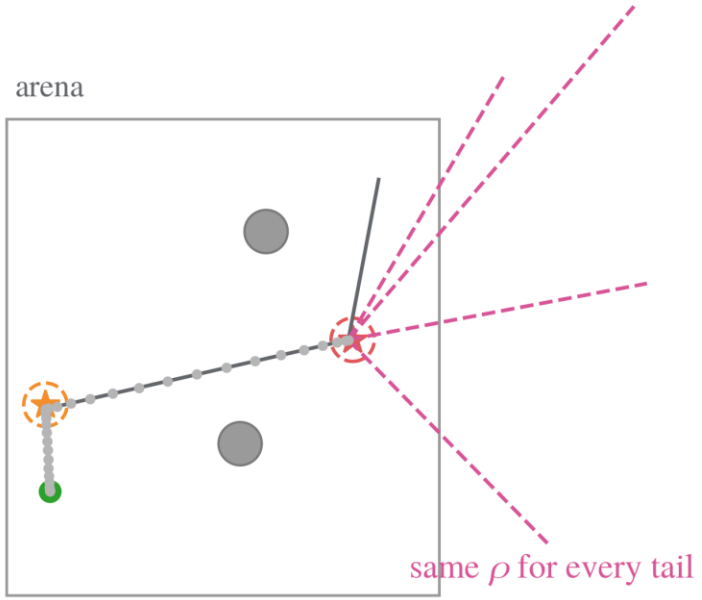
STL Optimizer for sequence with avoidance leaves the arena

Underdetermination by robustness

the score depends on a few waypoints



where only avoidance applies, the maximizers are unbounded



At most m of 64 to 256 waypoints determine the score, with m between 1 and 5

Robustness alone does not determine a trajectory (backup)

Cube manipulation

Nested reach-and-dwell

$$\varphi = \Diamond_{[0,30]}(\Box_{[0,10]}\, pick \ \wedge \ \Diamond_{[0,30]}\Box_{[0,10]}\, place)$$

Sequence

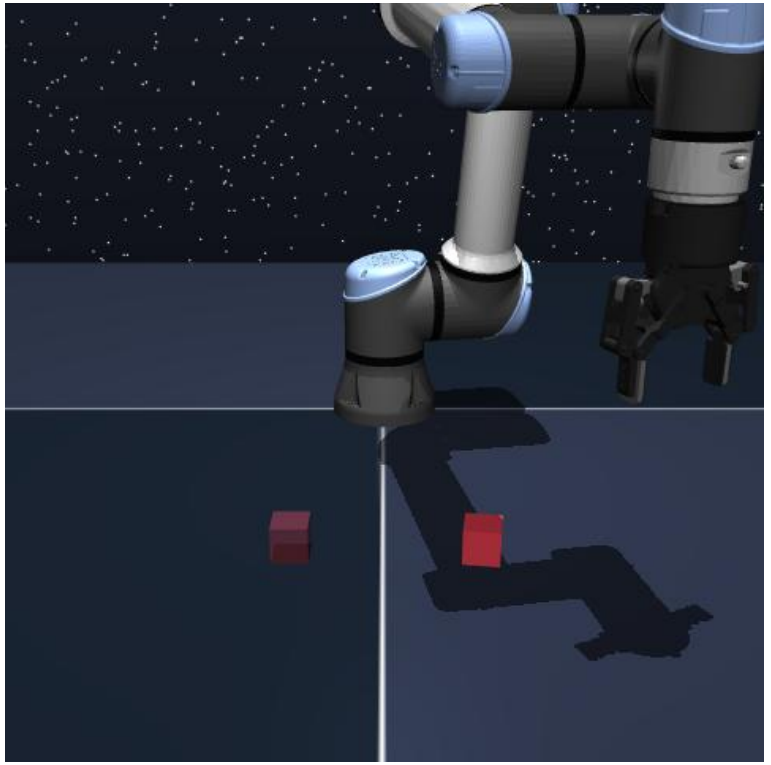
$$\varphi = \Diamond_{[0,20]}\, pick \ \wedge \ \Diamond_{[20,100]}\, place$$

Sequence with avoidance

$$\varphi = \Diamond_{[0,20]}\, pick \ \wedge \ \Diamond_{[20,100]}\, place \ \wedge \ \Box_{[0,127]}\, \neg obs$$

pick, the effector within 5 cm of the cube
place, the cube within 4 cm of the target, the environment's tolerance
obs, a 2 cm box 20 cm from the midpoint, which the cube must avoid

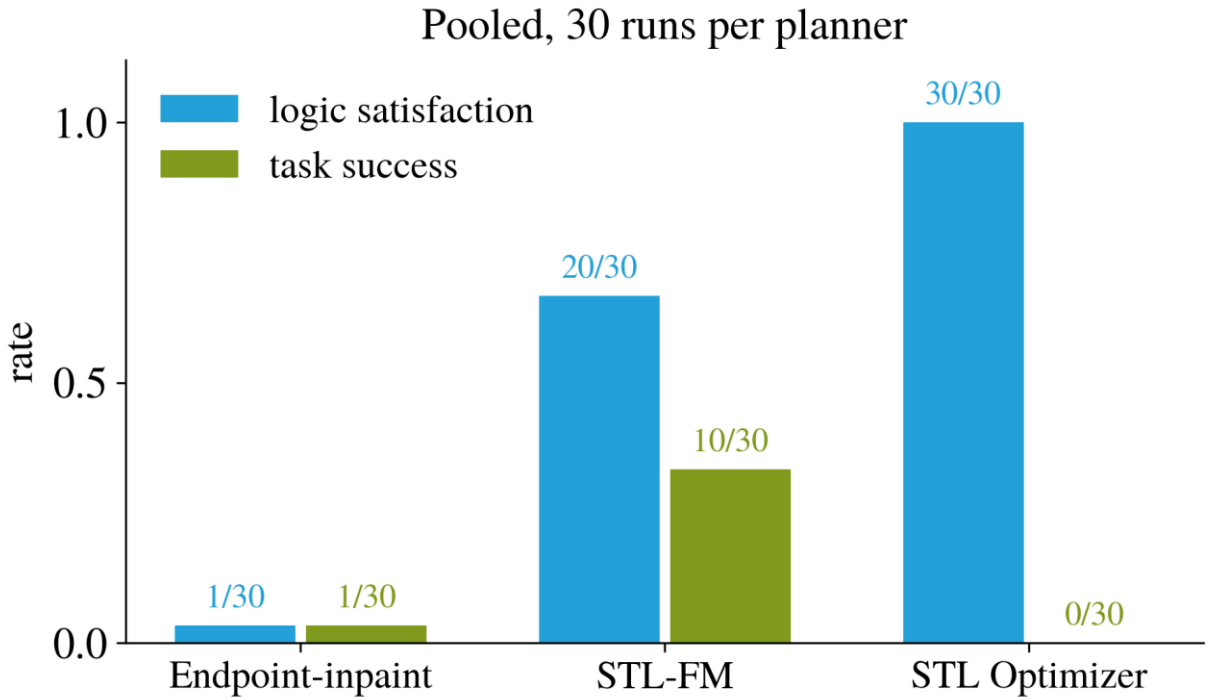
10 runs per class and planner, each plan executed in the simulator



STL-FM plan, executed, nested reach-and-dwell

The requirement and the environment's success criterion refer to the same cube

Logic satisfaction and task success



Logic satisfaction on the plan does not imply task success on execution

Conclusions

RQ1	**What does a demonstration witnessing a requirement imply about that requirement?** A demonstration witnesses a requirement without determining it, so the requirement is stated separately from the demonstrations.
RQ2	**When does maximizing the robustness score alone under-determine a trajectory, and when does it not?** The robustness score alone does not determine a trajectory. Without the prior, plans need additional specifications or constraints (staying leave the arena or the planned cube moves without contact)
Method	**STL-FM guides the learned prior at one constant strength, without retraining** STL-FM satisfies the dwell, sequence and avoidance requirements at rates of 0.78 to 1.00 per class and completes 10 of 30 cube executions, against none for the STL Optimizer.

Logic satisfaction and task success do not coincide. A requirement constrains only the variables the prior generates.

Future work

1. **A differentiable dynamics model**
   so that a requirement can constrain contact and object pose

2. **Execution on the physical platform and multimodal input**
   plans for the Franka Kitchen and GoFA door task, executed and scored on the robot

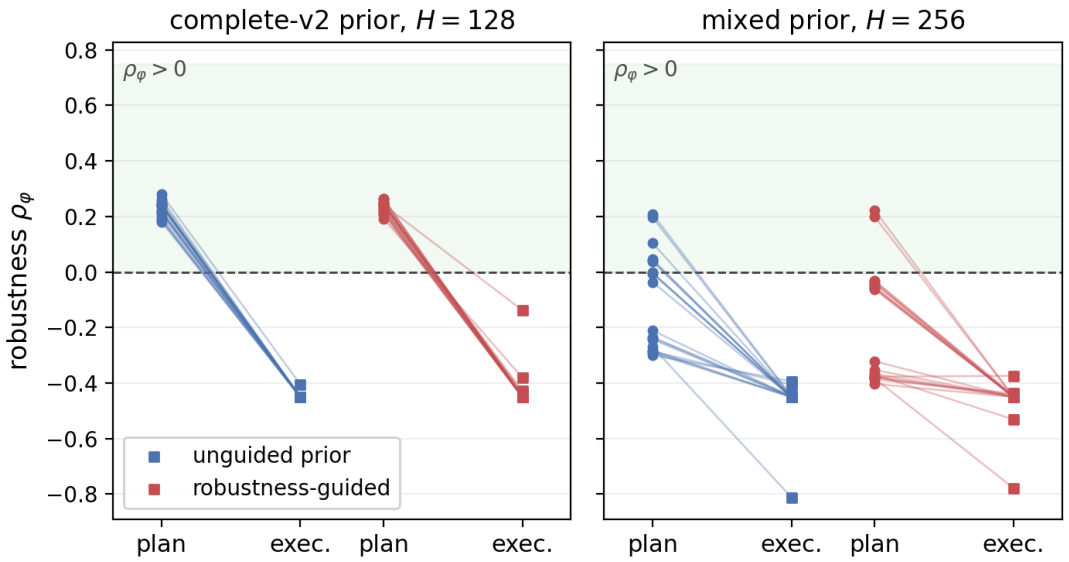
3. **A sweep of a requirement's parameters**
   the same prior under changed windows and dwell times

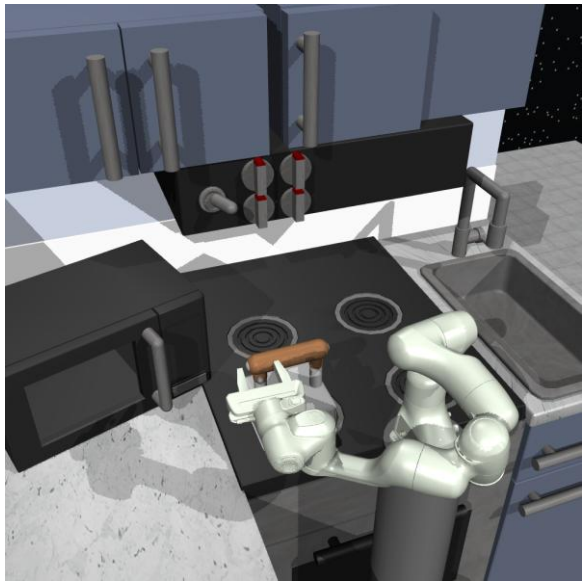
4. **A robust controller for execution**
   for an execution-time guarantee

5. **STL guidance for robot foundation models**
   VLAs and world action models already sample actions by flow matching

Franka Kitchen Benchmark

$$\varphi = \Diamond_{[60,130]}\, microwave \;\wedge\; \Diamond_{[140,230]}\, kettle$$





Complete-v2 prior, unguided,
mean robustness of plan +0.23
and of execution −0.45, a fall of 0.68
against a margin of at most 0.3

Task success in 0 of 80 executions

Physical platform



$$\varphi = \square_{[grasp,\,place]}(|\text{roll}|,\ |\text{pitch}|,\ |\text{yaw}| < 5^\circ)$$

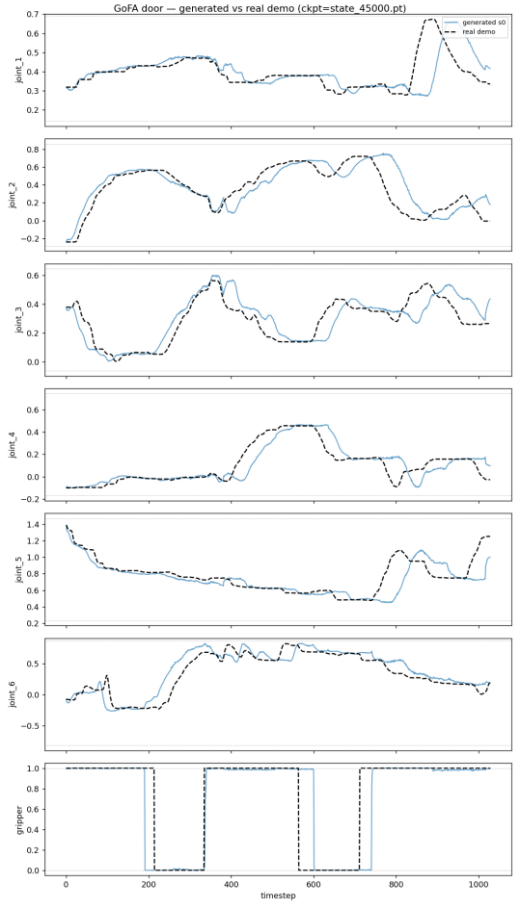


ABB GoFA CRB 15000,
five door-opening
demonstrations,
six joint angles and
the gripper state,
H = 128

No demonstration
satisfies the carry
requirement.

Dataset must be diverse.



Thank you!

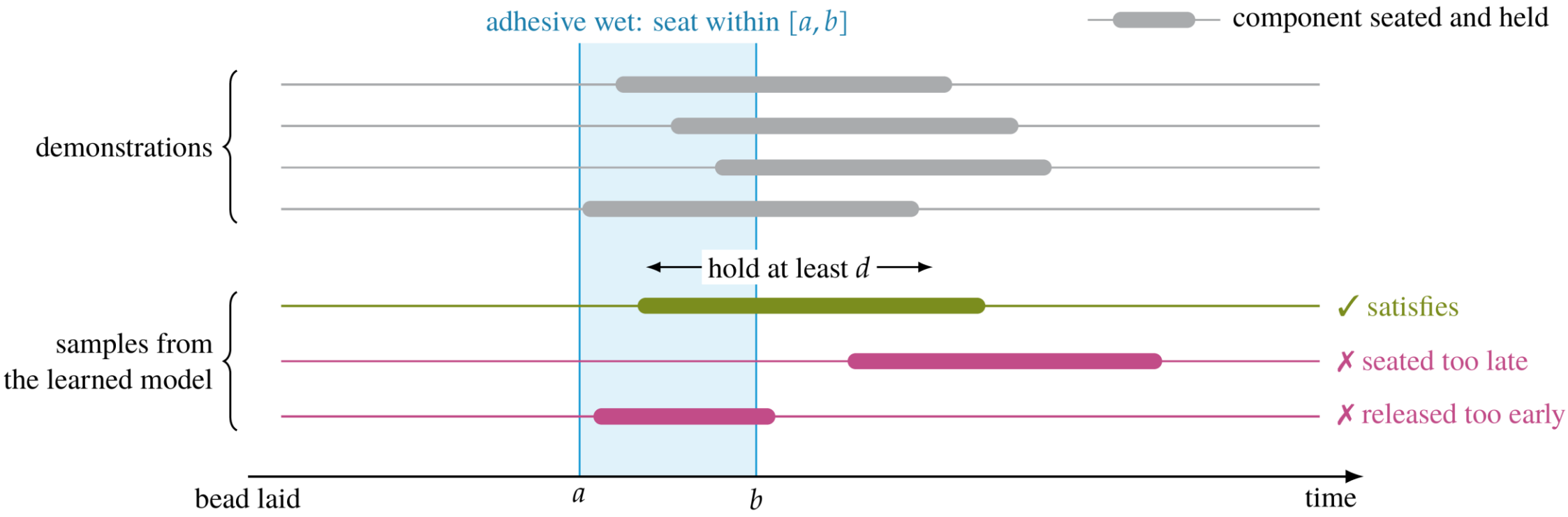
Open to Questions and Discussions


KTH
VETENSKAP
OCH KONST

Supplementary material

Answers to likely questions

Plausible samples that violate the requirement



Comparison of approaches

Approach	Requirement enters	Rest of the motion from	Property
Trajectory optimization Kurtz and Lin 2021	cost or constraint, with a known dynamics model	dynamics model, costs and feasibility constraints	specified by hand
Control barrier functions Lindemann and Dimarogonas 2019	time-varying barrier functions, quadratic program	dynamics model and control cost	boundedness conjoined to the requirement
Training-time conditioning TeLoGraF, Meng and Fan 2025	training, from requirement–demonstration pairs	conditioned flow-matching model	degrades outside the conditioning distribution
Decompose-and-stitch ZSTP, Liu et al. 2025	timed waypoints, before generation	pretrained diffusion model, between waypoints	timing of each operator fixed in advance
Safe-set enforcement SafeDiffuser, Xiao et al. 2023	barrier invariance in each denoising step	diffusion model	time-invariant safe set, no temporal operators
Inference-time robustness guidance CTG, Zhong et al. 2022	robustness gradient, during sampling	diffusion model	strength coupled to the noise level, no certificate
STL-FM this thesis	robustness gradient at the predicted endpoint, during sampling	learned prior, trained by flow matching	one constant strength, no retraining, no certificate

Entries summarize the cited works

Requirements witnessed by one demonstration

$$\mathcal{S}_\varphi = \{\tau \in \mathcal{T} : \tau \models \varphi\}, \qquad \{\tau^\star\} \subsetneq \mathcal{S}_{\varphi_1} \subsetneq \mathcal{S}_{\varphi_2} \subsetneq \cdots \subsetneq \mathcal{T}$$

A trajectory $\tau\star$ witnesses a requirement φ if it satisfies φ, that is, if $\tau\star$ lies in the satisfaction set $\mathcal{S}_\varphi$. Two requirements are equivalent if they have the same satisfaction set.

Any one trajectory witnesses uncountably many inequivalent requirements.

Their satisfaction sets form strictly increasing chains, from $\{\tau\star\}$, “do exactly this”, to the whole trajectory space $\mathcal{T}$, “do anything”.

Consequently, no map from a single demonstration to a requirement is determined by the demonstration.

This answers RQ1. Witnessing constrains membership in a satisfaction set but does not determine which requirement was intended, and the requirement is stated separately from the demonstrations, as an STL formula.

Sparsity of the robustness gradient

$$\pi_t(\nabla_\tau \rho_\varphi(\tau)) = 0 \ \text{ for } t \notin \mathcal{A}(\tau), \qquad |\mathcal{A}(\tau)| \le m, \qquad \dim N(\tau) \ge (H - m)\, d$$

The result concerns a formula φ with m predicate occurrences, scored with the exact semantics at a trajectory τ where every min and max selection is unique.

Near τ the score depends on at most m waypoints, the active set $\mathcal{A}(\tau)$.

The gradient is zero at every other waypoint, and the score is constant along a subspace $\mathrm{N}(\tau)$ whose dimension in the position coordinates is at least $(\mathrm{H} - \mathrm{m})\mathrm{d}$.

Gradient ascent on the score leaves every coordinate in $\mathrm{N}(\tau)$ at its initial value for as long as the active set is unchanged.

Here H is the horizon, d the dimension of the position coordinates, and π_t the waypoint at step t.

In the experiments m lies between 1 and 5, against horizons of 64 to 256.

Maximizers of a reach requirement

$$\varphi = \Diamond_{[a,b]}\, \mu, \qquad h_\mu(x) = r - \|x_{\text{pos}} - c\|$$

$$\mathcal{O}_\varphi = \arg\max_\tau \rho_\varphi(\tau) = \bigcup_{t \in [a,b]} \{\tau \in \mathcal{T} : x_t^{\text{pos}} = c\}$$

The reach requirement φ holds if the position lies within a ball of radius r about c at some step of $[a, b]$.

Its largest score r is attained by every trajectory that is at c at some step of $[a, b]$, whatever it does at the other steps.

The maximizing set $\mathcal{O}_\varphi$ is therefore a union of affine subspaces, each of dimension $(H - 1)d$ in the position coordinates.

It contains trajectories of arbitrarily large velocity and acceleration, and no rule internal to ρ_φ selects among them.

Maximality of the score implies nothing about plausibility.

Unbounded maximizers under avoidance

$$\varphi = \square_{[0,H-1]} \neg\mu, \qquad \rho_\varphi(\tau) = \min_t \left(\|x_t^{\text{pos}} - c\| - r\right), \qquad \sup_\tau \rho_\varphi(\tau) = +\infty$$

The predicate μ is an unsafe ball of radius r about c.

For the pure safety requirement $\square_{[0,H-1]} \neg\mu$ the score is unbounded above, so no maximizer exists.

With at least one reach conjunct the supremum is finite and attained.

At a step covered only by avoidance conjuncts, displacing the waypoint of a maximizer far from every unsafe centre leaves the score unchanged, so the set of maximizers is unbounded.

In neither case does the requirement bound the trajectory.

The coordinates that the requirement leaves free are determined by a component other than the requirement, in STL-FM the learned prior.

Maximization over plausible trajectories

$$\max_{\tau} \ \rho_{\varphi}(\tau) \quad \text{subject to} \quad \tau \in \mathcal{M}$$

The plausible trajectories are idealized as a compact C^1 submanifold $\mathcal{M}$ of the trajectory space $\mathcal{T}$, and the score is maximized on $\mathcal{M}$ only.

Existence. The maximum is attained on $\mathcal{M}$, even where the unconstrained supremum is unattained or infinite, as for pure safety. Compactness of $\mathcal{M}$ provides the bound that the requirement lacks.

Determinacy. For a generic embedding of $\mathcal{M}$ the maximizers on $\mathcal{M}$ are isolated, except where $\mathcal{M}$ is tangent to the flat directions. In $\mathcal{T}$ they are not isolated, as the sparsity and reach results show.

Plausibility. Every constrained maximizer lies in $\mathcal{M}$, so the implausible maximizers of a reach requirement are excluded whenever they lie off $\mathcal{M}$.

The learned prior satisfies neither hypothesis, so whether it provides the bound is measured in the experiments, not assumed.

Constant guidance strength

$$\nabla_x \log p_t(x) = \frac{\alpha_t \, \mathbb{E}[x_0 \mid x_t = x] - x}{\sigma_t^2} \qquad \text{diffusion score, grows like } 1/\sigma_t^2 \text{ as } t \to 0$$

$$\frac{d}{ds}\psi_s = \tau_1 - \tau_0 = O(1) \qquad \text{flow-matching target, uniformly in } s$$

$$\|v_{\text{guided}}(\tau, s)\| \le \|v_\theta(\tau, s)\| + \lambda_0 = O(1) \quad \text{with } \|\widetilde{\nabla}\| = 1$$

Both terms are of the same bounded order at every flow time, so the ratio of guidance to prior is $\Theta(\lambda_0)$ and a constant λ_0 is well posed without a schedule. $\lambda_0 = 50$ in navigation and cube, without retuning.

On a diffusion substrate the guidance weight is coupled to the noise schedule, $\lambda(t) \propto \sigma_t^k$ with $k \in \{1, 2\}$.

Compilation and smoothing bias

$$\max_i z_i \le \mathrm{smax}_\kappa(z) \le \max_i z_i + \frac{\log n}{\kappa}, \qquad \min_i z_i - \frac{\log n}{\kappa} \le \mathrm{smin}_\kappa(z) \le \min_i z_i$$

$$\rho_\varphi^{(\kappa)}(\tau) > L\,\kappa^{-1}\log n \implies \tau \models \varphi$$

A formula is compiled by a post-order traversal of its syntax tree, with no learned component.

A predicate returns its margin h at every step, and negation changes the sign.

Conjunction and always return smin, disjunction and eventually smax, and until a nested combination of both.

Over n arguments the surrogate lies within $\log n / \kappa$ of the exact extremum, so the compiled score converges to the exact score as κ grows.

At finite κ a compiled score above $L\kappa^{-1}\log n$ implies satisfaction, where L is the depth of the syntax tree and n the largest arity of its nodes.

Guidance uses $\kappa = 10$, and satisfaction is always scored with the exact semantics.

Steps of the guided sampler

$$\hat{\tau} = \tau + (1 - s)\, v_\theta(\tau, s), \qquad \tau \leftarrow \tau + \Delta s\, (v_\theta(\tau, s) + \lambda_0\, g)$$

The sampler takes the requirement score ρ_φ, the start state, the prior field v_θ, the guidance strength λ_0, the smoothness weight γ and the number of steps K.

1. τ is drawn from $\mathcal{N}(0, I)$, and its start state is pinned.
2. At each step $k = 0, \ldots, K - 1$, with $s = k\Delta s$ and $\Delta s = 1/K$, the prior velocity $v_\theta(\tau, s)$ is evaluated.
3. The predicted endpoint $\hat{\tau}$ is formed from the straight-line path.
4. The gradient g of the penalized objective is taken at $\hat{\tau}$, masked to the position coordinates and normalized.
5. The iterate is advanced by the prior step plus $\lambda_0\, g\, \Delta s$, and the start state is pinned again.

An optional clamp on the guidance increment is part of the sampler and is off in all reported runs.

Reproducing the STL Optimizer

Removes the learned prior and optimizes the robustness score directly, instead of sampling from a trained generative model

Controls, not states	A free control sequence u, one 2D step per time index, starts from small random noise
Single-integrator dynamics	p[t+1] = p[t] + u[t]·Δt turns the controls into the trajectory that gets scored
Same compiled score	The requirement is scored by the masked, differentiable ρϕ that STL-FM's guidance also uses (stljax / STLCG++)
Violation objective, no prior	Plain gradient descent on max(0, −ρϕ); a finite-difference smoothness penalty and a velocity limit stand in for a learned prior
Budget	1,500–6,000 gradient steps, with the smoothing sharpness annealed from soft to sharp

The robustness engine is shared with STL-FM; the optimization built around it, single-integrator dynamics and the violation penalty, is a standard direct-optimization baseline, not something STLCG++ itself proposes.

Navigation results, all four planners

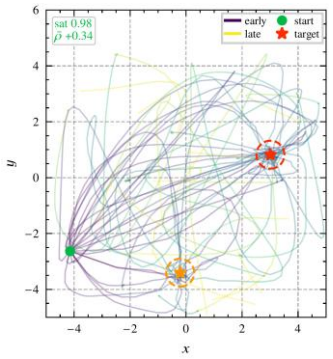
	(a) Endpoint-inpaint		(b) STL-FM (ours)		(c) STL-FM+pin (ours)		(d) Req.-only	
Class	sat	$\bar{\rho}$	sat	$\bar{\rho}$	sat	$\bar{\rho}$	sat	$\bar{\rho}$
1	**1.00**	**+0.50**	0.96	+0.34	**1.00**	**+0.50**	**1.00**	+0.13
2	0.00	−5.28	**1.00**	+0.38	**1.00**	**+0.39**	**1.00**	+0.03
3	0.00	−4.47	0.98	+0.29	0.99	**+0.30**	**1.00**	+0.03
4	0.00	−4.92	0.89	−0.04	0.87	−0.00	**1.00***	**+0.02***
5	0.00	−4.75	0.89	+0.04	0.94	**+0.06**	**1.00***	+0.03*
6	0.00	−5.34	0.78	+0.07	0.82	**+0.10**	**1.00**	+0.03
7	0.18	−1.09	**0.98**	**+0.34**	0.97	+0.32	**0.98**	+0.03
8	0.02	−3.57	0.96	**+0.21**	0.94	+0.17	**1.00***	+0.02*
9	0.05	−2.80	0.27	−0.08	**0.96**	**+0.24**	0.00*	−0.02*
10	0.03	−2.45	0.97	**+0.31**	0.98	**+0.31**	**1.00***	+0.05*

	(a)	(b) STL-FM			(c) STL-FM+pin			(d) Req.-only		
Class	s	first	best	s	first	best	s	first	best	s
1	0.21	30	94	0.42	2	2	0.64	15	27	5.67
2	–	89	98	0.59	89	100	0.82	9	18	10.58
3	–	84	98	0.63	84	98	0.92	4	10	26.94
4	–	86	95	0.75	86	97	1.02	23	41	13.95
5	–	83	97	0.85	84	97	1.05	20	35	28.13
6	–	94	100	1.15	92	98	1.36	24	34	28.22
7	0.23	37	95	0.54	41	95	0.75	26	50	23.02
8	–	84	97	0.75	84	97	0.96	31	57	22.92
9	0.20	83	95	5.72	47	94	5.86	–	–	–
10	0.22	53	95	4.69	56	95	4.87	22	49	32.03

Disjunction and until in navigation

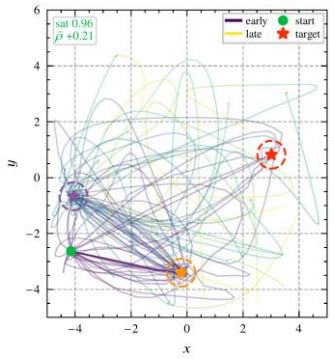
Choice of target

$\Diamond \mu_1 \vee \Diamond \mu_2$



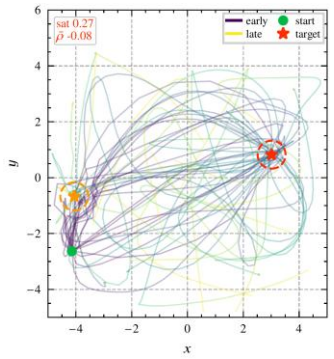
Choice of sequence

$(\Diamond \mu_1 \wedge \Diamond \mu_3) \vee (\Diamond \mu_2 \wedge \Diamond \mu_3)$



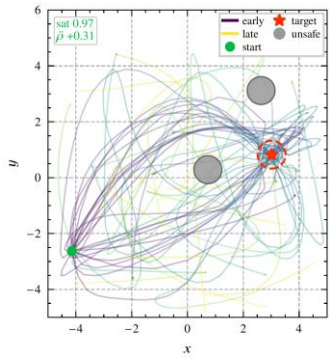
Ordered visit

$(\neg \mu_2 \, U \, \mu_1) \wedge \Diamond \mu_2$



Avoidance until reach

$(\neg \mu_3 \wedge \neg \mu_4) \, U \, \mu_1$

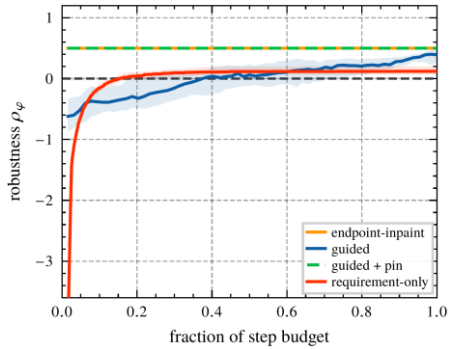


Class	Endpoint-inpaint	STL-FM	STL-FM with pin	STL Optimizer	Plan time, STL-FM	Plan time, STL Optimizer
Choice of target	0.18	0.98	0.97	0.98	0.54 s	23.02 s
Choice of sequence	0.02	0.96	0.94	1.00†	0.75 s	22.92 s
Ordered visit	0.05	0.27	0.96	0.00†	5.72 s	–
Avoidance until reach	0.03	0.97	0.98	1.00†	4.69 s	32.03 s

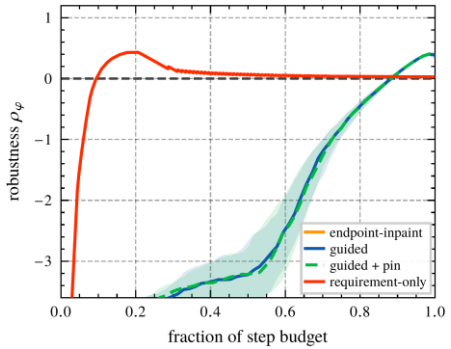
Satisfaction over 100 runs, and plan time as a median over satisfying timing runs. † every plan leaves the arena. The STL Optimizer fails ordered visit at a local optimum of its objective

Robustness against budget

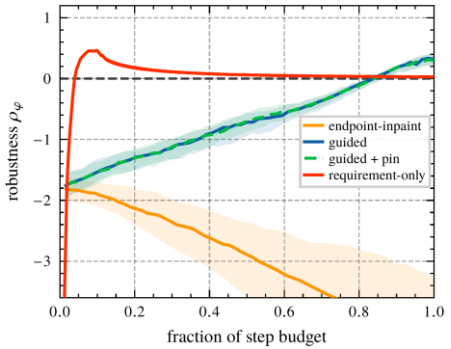
Reach



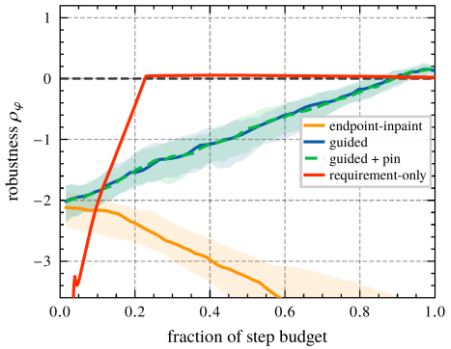
Nested reach-and-dwell



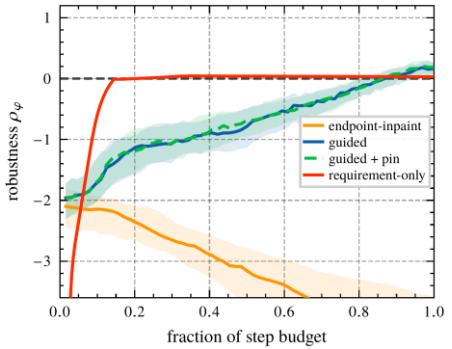
Recurrence



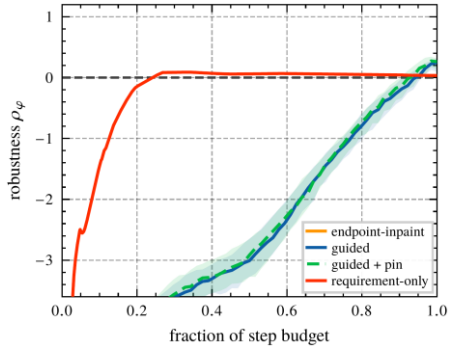
Sequence



Sequence with avoidance



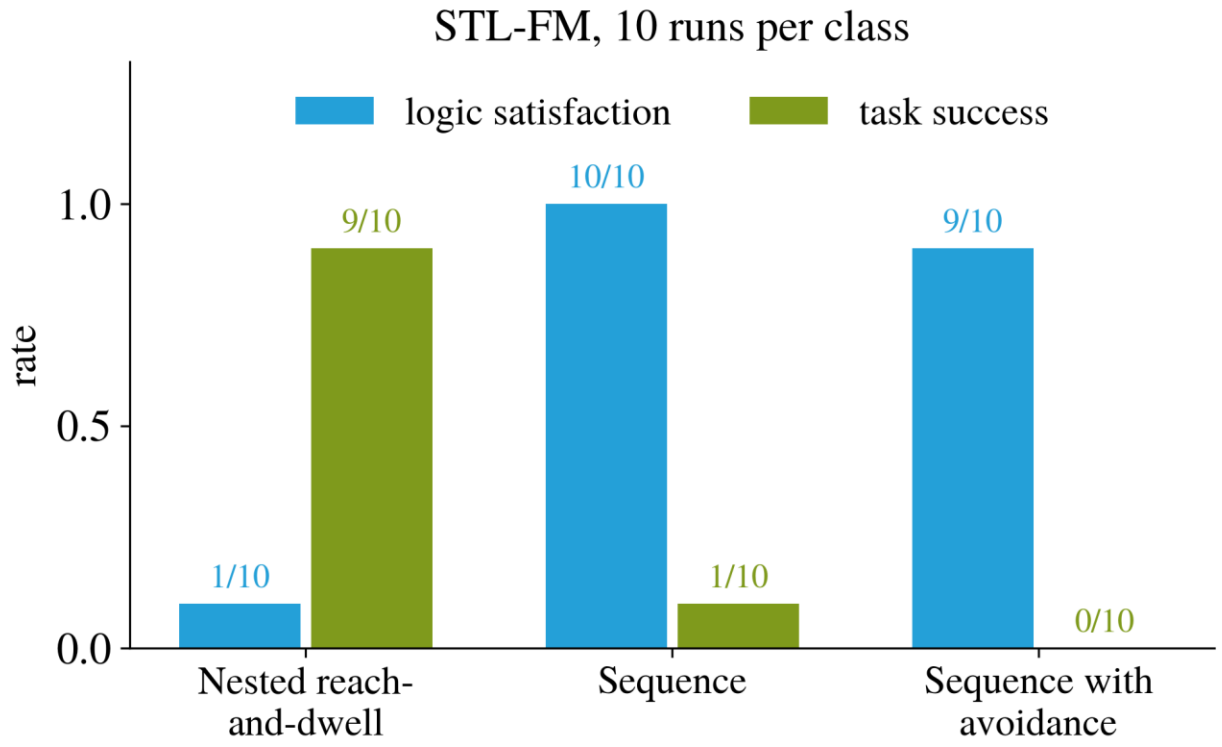
Composite



Cube results, all three classes

	(a) Endpoint-inpaint			(b) STL-FM			(d) Req.-only		
Class	sat	task	gap	sat	task	gap	sat	task	gap
2	0.00	0.00	5.7	0.10	0.90	7.0	1.00	0.00	7.3
4	0.00	0.00	6.4	1.00	0.10	6.8	1.00	0.00	13.0
5	0.10	0.10	7.4	0.90	0.00	6.7	1.00	0.00	13.6
Pooled	0.03	0.03	6.4	0.67	0.33	6.9	**1.00**	**0.00**	**12.5**

Logic satisfaction and task success per class



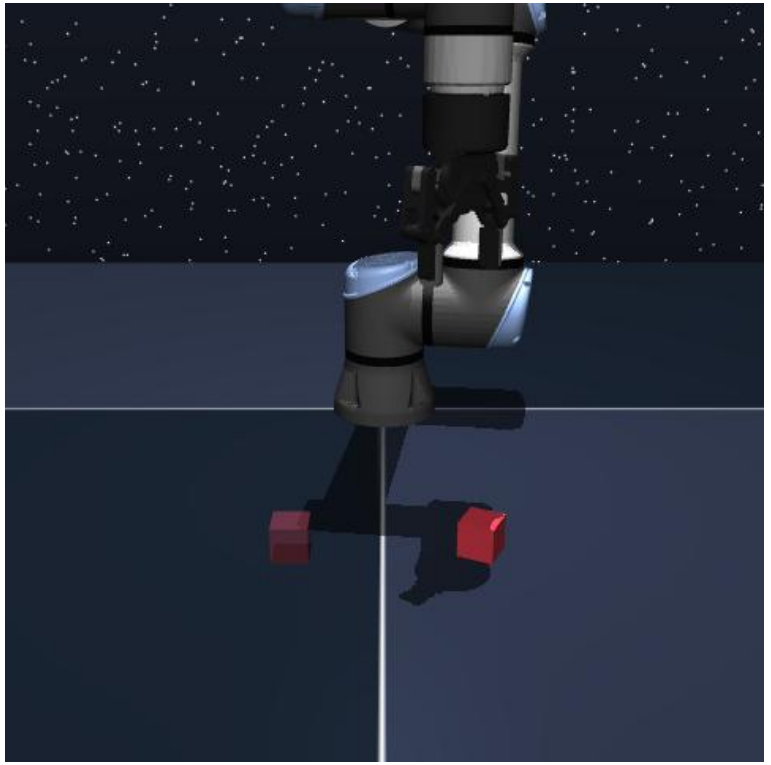
Cube manipulation, sequence

Sequence requirement

Grasp the cube within 20 steps,
place it between steps 20 and 100

$$\varphi = \Diamond_{[0,20]}\, pick \;\wedge\; \Diamond_{[20,100]}\, place$$

pick, the effector within 5 cm of the cube
place, the cube within 4 cm of the target, the environment's tolerance



STL-FM plan, executed

STL-FM satisfies the requirement in 10 of 10 plans and completes 1 execution

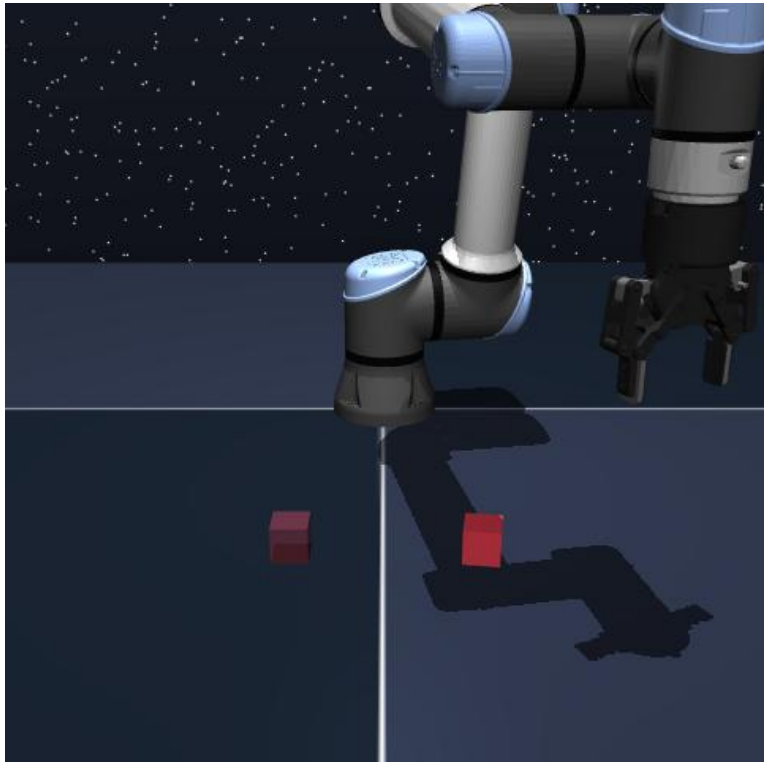
Cube manipulation, nested reach-and-dwell

Nested reach-and-dwell requirement

Grasp within 30 steps and hold for 10,
then place within 30 and hold for 10

$$\varphi = \Diamond_{[0,30]}(\Box_{[0,10]}\, pick \ \wedge \ \Diamond_{[0,30]}\Box_{[0,10]}\, place)$$

pick, the effector within 5 cm of the cube
place, the cube within 4 cm of the target, the environment's tolerance



STL-FM plan, executed

STL-FM completes 9 of 10 executions and satisfies the requirement in 1

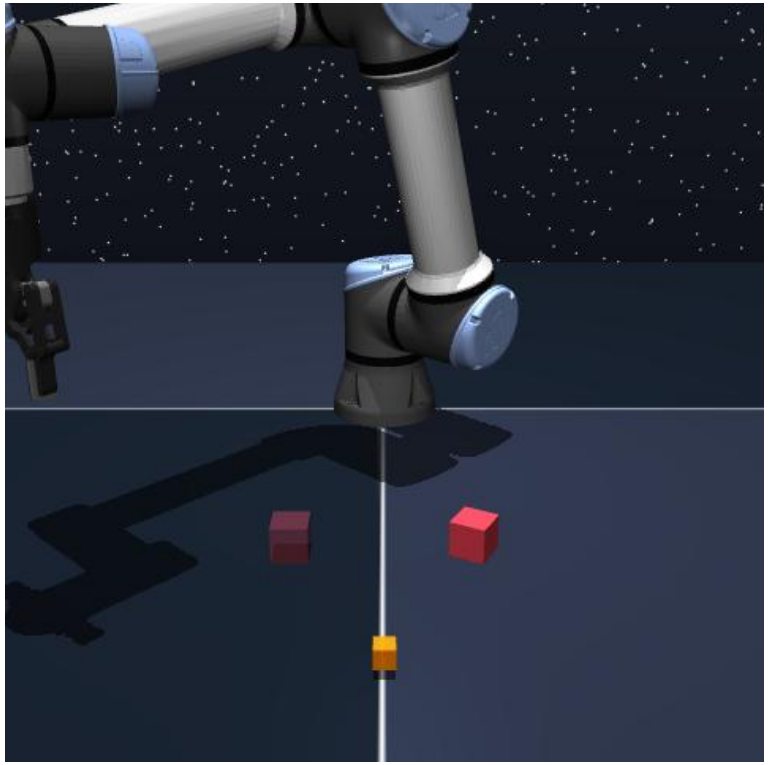
Cube manipulation, sequence with avoidance

Sequence-with-avoidance requirement

Grasp and place as in the sequence,
keeping the cube clear of the obstacle

$$\varphi = \Diamond_{[0,20]}\, pick \;\wedge\; \Diamond_{[20,100]}\, place \;\wedge\; \Box_{[0,127]}\, \neg obs$$

pick and place as above; obs, a 2 cm box 20 cm from the midpoint of pick and place, which the cube must avoid



STL-FM plan, executed

STL-FM satisfies the requirement in 9 of 10 plans and completes no execution

Roughness of guided plans

Trajectories	Median turn angle	Turns above 90°
Demonstrations	0.6°	2 %
Unguided prior, start pinned	6.4°	4 %
Unguided prior, both endpoints pinned	8.2°	7 %
STL-FM, nested reach-and-dwell	30.9°	47 %
STL-FM, recurrence	9.9°	
STL-FM, composite		18.5 %
STL Optimizer, pooled	0.3°	

The evaluation does not distinguish two explanations. Either the prior is undertrained, or guidance moves iterates away from plausible trajectories faster than the prior moves them back. The evidence for the contribution of the prior is the carry gap and the arena excursion.

Limitations

Fixed requirement instances	every rate is measured on one instance per class
No certificate	the guided sampler is a soft realization, so it gives a rate, not a guarantee
Sampled-state semantics	robustness does not constrain the path between two sampled states
Requirement-neutral prior	a modelling assumption, although demonstrations can retain task structure
Reproducibility	runs with identical seeds agree on the satisfaction count in 27 of 28 cells and differ by one run in the other
Physical carry interval	a proxy read from the gripper trace, longer than the planning horizon